

The Numerical Range Is a 2-Spectral Set

A Semester Workshop from First Principles

Pedagogical notes based on the manuscript
The numerical range is a 2-spectral set by Shanmu Jin

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Preface

These notes turn the proof in the source manuscript [1] into a semester-long workshop. The intended reader does *not* need prior exposure to Crouzeix’s conjecture, numerical ranges, spectral sets, positive kernels, double-layer potentials, or matrix-valued Herglotz theory. The phrase “from ABC/101” here means that no problem-specific vocabulary or folklore is assumed. A first course in finite-dimensional linear algebra and the basic definitions of complex analysis are useful. The notes review the needed objects from the ground up and isolate deeper classical results—such as Schur triangularization, Cauchy’s formula, and Runge approximation—in clearly marked boxes with proof blueprints.

The central theorem is easy to state. For a complex square matrix A , define its numerical range by

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}.$$

For every complex polynomial p , the source manuscript presents a proof of

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|.$$

The Week 3 nilpotent example shows that no valid universal constant can be smaller than 2; thus the manuscript’s claimed bound would be sharp.

What makes the proof worth a workshop is not only the conclusion. The new idea is a reusable two-stage mechanism:

structured cancellation in a positive kernel $\implies$ rigidity of two weighted Gramians.

An unknown analytic correction is not bounded or discarded; a carefully chosen sample at the origin cancels it *exactly*. The surviving inequality is converted into the sharp constant by a Stein identity. The middle third of the course studies this mechanism line by line.

Status and purpose of these notes

These notes are a pedagogical expansion of the stated source manuscript, which the repository describes as a candidate proof with formal peer review pending. They are not an independent refereeing report. They expose every logical dependency so that a workshop can check the argument rather than merely quote it. When a classical theorem such as Runge’s theorem or the scalar Herglotz representation is used, its precise statement, role, and proof blueprint are supplied.

How to use the material

Each numbered chapter is one workshop week. A typical meeting can use two hours for guided theory and one hour for problems. The boxes have consistent roles:

- **Learning goals** say what a participant should be able to do by the end of the week.
- **Intuition** translates a formal step into a mental model.
- **Checkpoint** pauses the proof and records exactly what has been earned.
- **Common pitfall** marks a tempting but invalid shortcut.
- **Classical input** isolates a theorem whose full proof belongs to a standard prerequisite course.

Exercises are marked *warm-up*, *core*, or *stretch*. Hints and selected solutions appear in [Chapter B](#). A reader following only the core route should complete all core exercises and the proof-reconstruction checklists in Weeks 12 and 14.

A short Week 0 diagnostic

Before Week 1, check that you can multiply 2×2 complex matrices, solve a two-equation linear system, and differentiate z^k . Everything else needed for the argument is either developed in the notes or labeled as a classical input. If the words “spectrum,” “holomorphic,” or “positive semidefinite” are new, that is expected: Weeks 1, 2, and 5 introduce them.

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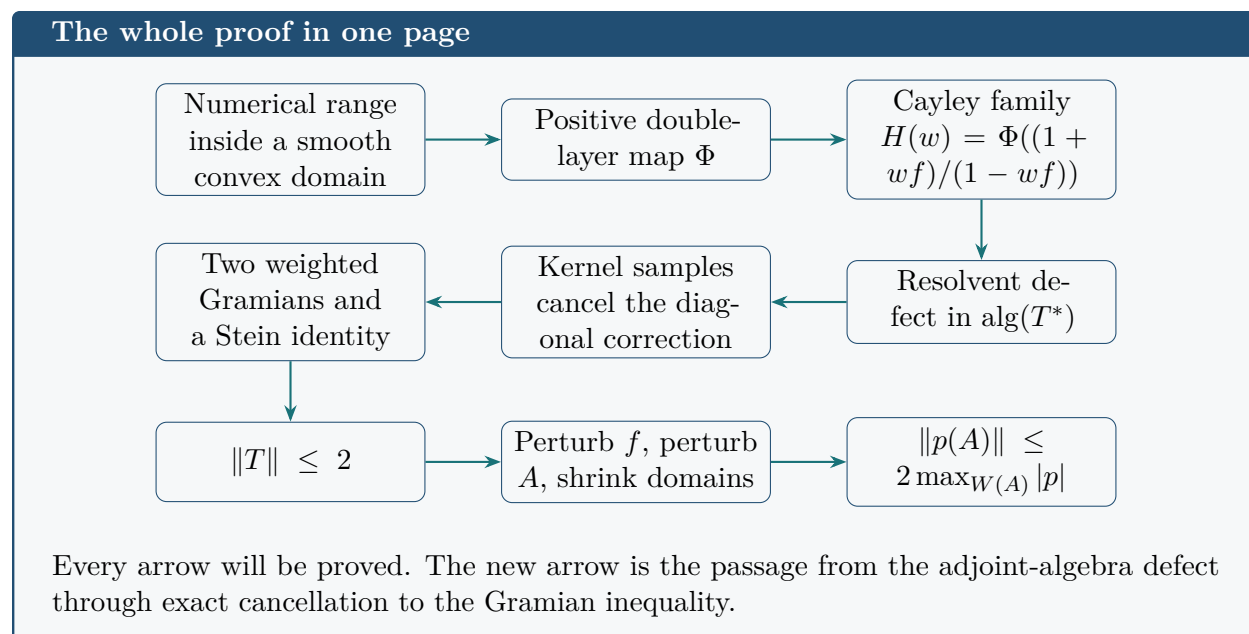
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Semester road map

Week	Theme	Proof milestone	Deliverable
1	Matrix language	Norms, adjoints, positivity, and congruence	Fluency sheet
2	Spectra and matrix functions	Generated algebras and interpolation	Functional-calculus map
3	Numerical ranges	Geometry, convexity, and sharp 2×2 example	Geometry problem set
4	Spectral sets	The conjecture and normalization	One-page problem brief
5	Complex and convex analysis	Resolvents, Cauchy calculus, smooth boundaries	Contour-calculus sheet
6	Positive maps and measures	The language behind the boundary calculus	Positivity audit
7	Herglotz kernels	Positive-real functions become matrix inequalities	Kernel sampling lab
8	Double-layer calculus	Construct the unital positive map Φ	Derivation portfolio
9	Cayley completion	Produce H and locate its resolvent defect	Completion proof
10	Diagonal algebra and cancellation	Diagonalize the defect and cancel it with half-eigenvalue and origin samples	Change-of-basis audit
11	Surviving inequality	Compute the known blocks and identify two Gramians	Cancellation calculation
12	Gramian rigidity	Two scales, an eigenvector test, and a Stein identity	Core theorem oral proof
13	Removing hypotheses	Three approximation layers	Limit ledger
14	Main theorem	Assemble the polynomial and rational conclusions	Full proof write-up
15	Consequences and computation	Holomorphic calculus, compression, algorithms	Numerical experiment
16	New-tool clinic	General scale, adjoint alignment, and open directions	Capstone presentation



Deliverables and a common rubric

The final column of the road map names one concrete artifact per week. A “sheet” is a two-page collection of definitions and one checked example; an “audit” annotates every change of basis or limiting step; a “portfolio” collects the week’s guided derivations; and a “proof” is a dependency-ordered argument with no unexplained symbols. The numerical experiment and capstone may be completed in pairs; all other artifacts are individual.

Each deliverable is assessed on the same four questions, one point each:

1. Are all objects defined before use?
2. Is every equality justified by a named fact or a displayed calculation?
3. Are positivity, convergence, and change-of-basis claims made in the correct direction?
4. Can the author state in two sentences how the artifact advances the main proof?

A score of 3/4 is ready for the next week; a score of 2/4 or below should be revised using the corresponding hints in [Chapter B](#). For the Week 12 oral proof and Week 14 full write-up, all four criteria are required.

Part I

The objects in the statement

Chapter 1

Matrix language: vectors, norms, adjoints, and positivity

Learning goals

By the end of this week, you should be able to:

- compute with the complex inner product and conjugate transpose;
- distinguish vector, operator, and scalar norms;
- translate a positive-semidefinite matrix inequality into inequalities of quadratic forms;
- explain why unitary similarity and congruence play different roles in the proof.

Workshop plan

Opening (20 min): compare Ax , x^*Ax , and eigenvalues. **Core theory (70 min):** norms, adjoints, Hermitian matrices, positive semidefiniteness. **Guided proof (35 min):** congruence and square roots. **Problem studio (45 min):** exercises 1.2 to 1.4.

1.1 Complex vectors and the adjoint

A vector in $\mathbb{C}^n$ is a column $x = (x_1, \dots, x_n)^T$. Its conjugate transpose is

$$x^* = (\overline{x_1}, \dots, \overline{x_n}).$$

We use the inner product and Euclidean norm

$$\langle x, y \rangle = x^*y = \sum_{j=1}^n \overline{x_j}y_j, \quad \|x\|_2 = \sqrt{x^*x}.$$

This convention is conjugate-linear in the first variable and linear in the second. For a matrix $A = (a_{ij})$, the adjoint is $A^* = \overline{A}^T$, characterized by

$$\langle Ax, y \rangle = \langle x, A^*y \rangle.$$

Definition 1.1. A matrix $U \in \mathbb{M}_n(\mathbb{C})$ is *unitary* if $U^*U = UU^* = I$. A matrix H is *Hermitian* if $H = H^*$. A matrix A is *normal* if $A^*A = AA^*$.

Unitary matrices preserve inner products and lengths. They are the complex analogue of orthogonal changes of coordinates. The finite-dimensional spectral theorem says that A is normal if and only if it is unitarily diagonalizable. Hermitian and unitary matrices are normal; upper-triangular matrices with a nonzero off-diagonal entry are typical nonnormal examples.

1.2 The induced operator norm

For $A \in \mathbb{M}_n(\mathbb{C})$, its induced Euclidean operator norm is

$$\|A\| = \max_{\|x\|_2=1} \|Ax\|_2.$$

The maximum exists because the unit sphere is compact and the norm is continuous. Three facts will be used repeatedly:

$$\|Ax\|_2 \leq \|A\| \|x\|_2, \quad (1.1)$$

$$\|AB\| \leq \|A\| \|B\|, \quad (1.2)$$

$$\|UAV\| = \|A\| \quad (U, V \text{ unitary}). \quad (1.3)$$

Moreover,

$$\|A\|^2 = \lambda_{\max}(A^*A),$$

where $\lambda_{\max}$ is the largest eigenvalue of a Hermitian matrix. Thus $\|A\|$ is the largest singular value, not in general the largest modulus of an eigenvalue.

Intuition

Eigenvalues measure what A does on its invariant directions. The operator norm asks for the largest stretch in *any* direction. For a nonnormal matrix, those can be very different. Crouzeix's problem is fundamentally about controlling this nonnormal stretch.

1.3 Positive-semidefinite order

Definition 1.2. A Hermitian matrix H is *positive semidefinite*, written $H \succeq 0$, if

$$x^* H x \geq 0 \quad \text{for every } x \in \mathbb{C}^n.$$

For Hermitian H, K , write $H \preceq K$ when $K - H \succeq 0$.

The spectral theorem gives the equivalent conditions

$$H \succeq 0 \iff \text{every eigenvalue of } H \text{ is nonnegative} \iff H = R^* R \text{ for some } R.$$

Every positive-definite $G \succ 0$ has a unique positive-definite square root $G^{1/2}$, and $G^{-1/2}$ is well defined.

Lemma 1.3 (Congruence preserves positivity). *If $H \succeq 0$, then $S^* H S \succeq 0$ for every matrix S . If S is invertible, the converse also holds.*

Proof. For every x ,

$$x^* S^* H S x = (Sx)^* H (Sx) \geq 0.$$

When S is invertible, apply the same argument with S^{-1} . □

Similarity $S^{-1} A S$ preserves eigenvalues. Congruence $S^* H S$ preserves positivity. They coincide only for a unitary S ; mixing them up is a recurring source of errors.

1.4 Real and imaginary parts of a matrix

Define

$$\operatorname{Re} A = \frac{A + A^*}{2}, \quad \operatorname{Im} A = \frac{A - A^*}{2i}.$$

Both are Hermitian and $A = \operatorname{Re} A + i \operatorname{Im} A$. The expression $\operatorname{Re} A \succeq 0$ means the Hermitian part of A is positive; it does not assert that A itself is Hermitian.

Lemma 1.4 (Zero quadratic form for a positive matrix). *If $H \succeq 0$ and $x^* H x = 0$, then $Hx = 0$.*

Proof. Write $H = H^{1/2} H^{1/2}$. Then $x^* H x = \|H^{1/2} x\|_2^2 = 0$, so $H^{1/2} x = 0$, hence $Hx = 0$. $\square$

This tiny lemma is decisive in Week 12: equality in a positive Gramian will force a vector into a kernel.

1.5 Exercises

Exercise 1.1 (Warm-up: adjoints). Verify $(AB)^* = B^* A^*$, $(A^{-1})^* = (A^*)^{-1}$, and $\|A^*\| = \|A\|$.

Exercise 1.2 (Core: the sharp example). For $N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, compute $N^* N$ and show $\|N\| = 2$.

Exercise 1.3 (Core: positivity tests). For a Hermitian matrix $H = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$, prove that $H \succeq 0$ if and only if $a \geq 0$, $d \geq 0$, and $ad \geq |b|^2$.

Exercise 1.4 (Core: congruence versus similarity). Give a 2×2 example showing that similarity need not preserve Hermitianity or positive semidefiniteness. Then prove [theorem 1.3](#) without referring to eigenvalues.

Exercise 1.5 (Stretch: order and norm). For Hermitian H , prove $-cI \preceq H \preceq cI$ if and only if $\|H\| \leq c$.

Chapter 2

Spectra, diagonalization, and functions of a matrix

Learning goals

- Define spectrum, resolvent, and polynomial evaluation.
- Explain why distinct eigenvalues imply diagonalizability.
- Use Lagrange interpolation to identify $\text{alg}(T)$.
- Recognize the holomorphic functional calculus, which Week 5 develops in analytic detail.

Workshop plan

Opening: why $p(A)$ contains more information than $p(\sigma(A))$. **Core:** spectrum, diagonalization, generated algebras. **Guided proof:** Lagrange interpolation. **Studio:** calculate functions of diagonal and Jordan matrices.

2.1 Eigenvalues, spectrum, and resolvent

A number $\lambda \in \mathbb{C}$ is an eigenvalue of A if $Ax = \lambda x$ for some nonzero vector x . The spectrum is

$$\sigma(A) = \{\lambda \in \mathbb{C} : \lambda I - A \text{ is not invertible}\}.$$

In finite dimensions this is exactly the set of roots of $\det(\lambda I - A)$. When $z \notin \sigma(A)$, the matrix

$$R_A(z) = (zI - A)^{-1}$$

is the *resolvent*.

The spectral radius is

$$\rho(A) = \max_{\lambda \in \sigma(A)} |\lambda|.$$

Since $Ax = \lambda x$ for a unit eigenvector, $|\lambda| \leq \|A\|$, so $\rho(A) \leq \|A\|$. Equality need not hold.

2.2 Polynomial evaluation

If $p(z) = a_0 + a_1 z + \cdots + a_d z^d$, define

$$p(A) = a_0 I + a_1 A + \cdots + a_d A^d.$$

This definition respects sums and products:

$$(p + q)(A) = p(A) + q(A), \quad (pq)(A) = p(A)q(A).$$

The spectral mapping theorem says

$$\sigma(p(A)) = p(\sigma(A)).$$

It controls the eigenvalues of $p(A)$, but Crouzeix's problem asks for the generally larger quantity $\|p(A)\|$.

Example 2.1 (A nilpotent warning). For N from [exercise 1.2](#), $\sigma(N) = \{0\}$ and $\rho(N) = 0$, yet $\|N\| = 2$. The eigenvalues alone give no useful norm bound.

2.3 Diagonalization and simple spectrum

If A has n linearly independent eigenvectors, place them in the columns of an invertible matrix S . Then

$$A = S\Lambda S^{-1}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Consequently,

$$p(A) = S \text{diag}(p(\lambda_1), \dots, p(\lambda_n)) S^{-1}.$$

Eigenvectors belonging to distinct eigenvalues are linearly independent, so a matrix with n distinct eigenvalues is diagonalizable. This condition is called *simple spectrum*.

Common pitfall

The diagonalizing matrix S need not be unitary. Therefore $\|S\Lambda S^{-1}\|$ can be much larger than $\|\Lambda\|$. Replacing a nonnormal matrix by its diagonal eigenvalue list loses the geometry stored in S^*S .

2.4 The unital algebra generated by a matrix

Definition 2.2. The unital algebra generated by T is

$$\text{alg}(T) = \{q(T) : q \in \mathbb{C}[z]\}.$$

By Cayley–Hamilton, every high power of T is a linear combination of $I, T, \dots, T^{n-1}$, so $\text{alg}(T)$ is finite-dimensional and therefore norm closed.

Lemma 2.3 (Interpolation description). *Suppose*

$$T = S \text{diag}(\lambda_1, \dots, \lambda_n) S^{-1}$$

with distinct λ_j . Then

$$\text{alg}(T) = \{SDS^{-1} : D \text{ diagonal}\}.$$

Moreover,

$$\text{alg}(T^*) = \{(S^{-1})^* D S^* : D \text{ diagonal}\}.$$

Proof. Every polynomial in T has the first displayed form. Conversely, given $D = \text{diag}(d_1, \dots, d_n)$, Lagrange interpolation gives a polynomial q with $q(\lambda_j) = d_j$, hence $q(T) = SDS^{-1}$.

Taking adjoints in $T = SAS^{-1}$ gives

$$T^* = (S^{-1})^* \bar{A} S^*.$$

The conjugate eigenvalues remain distinct, so the same interpolation argument proves the second formula. $\square$

The particular orientation of the second formula will drive the exact cancellation in Week 10.

2.5 First preview of the holomorphic functional calculus

Let f be holomorphic on an open set containing $\sigma(A)$. Because the spectrum is finite, choose disjoint small closed disks inside that open set whose interiors contain all the distinct spectral points, and let Γ be the union of their positively oriented boundary circles. Define

$$f(A) = \frac{1}{2\pi i} \int_{\Gamma} f(z)(zI - A)^{-1} dz. \quad (2.1)$$

For polynomials this agrees with the earlier definition. If A is diagonalizable, it gives

$$f(A) = S \text{diag}(f(\lambda_1), \dots, f(\lambda_n)) S^{-1}.$$

For a nondiagonalizable matrix, derivatives of f at eigenvalues appear; Hermite interpolation supplies a polynomial q with $f(A) = q(A)$. In particular,

$$f(A) \in \text{alg}(A) \quad \text{whenever } f \text{ is holomorphic near } \sigma(A). \quad (2.2)$$

This is a first-pass algebraic preview. Week 5 defines holomorphic functions and oriented contour integrals, states Cauchy's formula, and explains why the definition is independent of the chosen circles. For the Week 2 exercises, it is enough to use the displayed formula or the diagonal/Jordan formulas that follow from it.

Checkpoint

Weeks 9–10 will produce matrices $g_{f^m}(B)$ by contour integration. Equation (2.2) is the reason they lie in $\text{alg}(B)$; taking adjoints then places the unknown correction in $\text{alg}(B^*)$.

2.6 Neumann series beyond $\|X\| < 1$

The elementary geometric series says $(I - X)^{-1} = \sum_{k \geq 0} X^k$ when $\|X\| < 1$. In finite dimensions it also converges whenever $\rho(X) < 1$. One way to see this is to use Jordan form: a Jordan block contributes a polynomial in k times $|\lambda|^k$, which still tends to zero and is summable when $|\lambda| < 1$. Therefore, if $\sigma(T) \subset \overline{\mathbb{D}}$ and $w \in \mathbb{D}$,

$$(I - wT)^{-1} = \sum_{m=0}^{\infty} w^m T^m \quad \text{in operator norm.} \quad (2.3)$$

2.7 Exercises

Exercise 2.1 (Warm-up). Compute $p(A)$ for $A = \text{diag}(1, i, -1)$ and $p(z) = z^3 - 2z + 1$.

Exercise 2.2 (Core: interpolation). Write the Lagrange polynomial taking prescribed values d_j at distinct points λ_j , and use it to prove [theorem 2.3](#).

Exercise 2.3 (Core: same generated algebra). Let B have simple spectrum $\beta_1, \dots, \beta_n$, and let f be holomorphic near $\sigma(B)$. If $f(\beta_1), \dots, f(\beta_n)$ are distinct, prove $\text{alg}(f(B)) = \text{alg}(B)$.

Exercise 2.4 (Core: Jordan calculus). For $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$, use [\(2.1\)](#) or a power series to show

$$f(J) = \begin{pmatrix} f(\lambda) & f'(\lambda) \\ 0 & f(\lambda) \end{pmatrix}.$$

Exercise 2.5 (Stretch: resolvent series). Prove [\(2.3\)](#) directly from Jordan normal form, including the case of a nontrivial Jordan block.

Chapter 3

The numerical range: geometry attached to a matrix

Learning goals

- Interpret x^*Ax as a directional scalar observation of A .
- Prove compactness, spectral containment, and unitary invariance.
- Understand why $W(A)$ is convex.
- Read exposed faces from maximal eigenspaces of rotated Hermitian parts.
- Compute the sharp nilpotent example.

Workshop plan

Opening: sample x^*Ax numerically for a 2×2 matrix. **Core:** basic properties and the 2×2 ellipse. **Guided proof:** Toeplitz–Hausdorff by compression. **Studio:** exact numerical ranges of elementary matrices.

3.1 Definition and first properties

Definition 3.1. For $A \in \mathbb{M}_n(\mathbb{C})$, its numerical range is

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n, \|x\|_2 = 1\}.$$

The scalar x^*Ax is the response of A observed in the same unit direction x . The following facts are immediate but important.

Proposition 3.2. For $A \in \mathbb{M}_n(\mathbb{C})$:

1. $W(A)$ is compact.
2. $\sigma(A) \subset W(A)$.
3. $W(U^*AU) = W(A)$ for every unitary U .
4. $W(\alpha A + \beta I) = \alpha W(A) + \beta$ for $\alpha, \beta \in \mathbb{C}$.
5. $W(A^*) = \overline{W(A)}$.

Proof. The unit sphere is compact and $x \mapsto x^*Ax$ is continuous, proving (1). A normalized eigenvector gives (2). Substitution $y = Ux$ proves (3), and direct calculation proves (4)–(5). $\square$

3.2 The 2×2 ellipse

Theorem 3.3 (Elliptical range theorem in dimension two). *For a 2×2 complex matrix B , $W(B)$ is a filled ellipse, possibly degenerating to a disk, a line segment, or a point. Its foci are the two eigenvalues of B .*

Guided calculation. Unitary Schur triangularization and a harmless phase change reduce to

$$B = \begin{pmatrix} \lambda_1 & c \\ 0 & \lambda_2 \end{pmatrix}, \quad c \geq 0.$$

Every unit vector, up to a scalar phase that cancels from x^*Bx , has the form

$$x = \begin{pmatrix} \cos t \\ e^{i\phi} \sin t \end{pmatrix}, \quad 0 \leq t \leq \frac{\pi}{2}.$$

Thus

$$x^*Bx = \frac{\lambda_1 + \lambda_2}{2} + \frac{\lambda_1 - \lambda_2}{2} \cos(2t) + \frac{c}{2} e^{i\phi} \sin(2t). \quad (3.1)$$

After translating the center to 0 and rotating $\lambda_1 - \lambda_2$ onto the real axis, put

$$s = \cos(2t), \quad u = \sin(2t) \cos \phi, \quad v = \sin(2t) \sin \phi.$$

Then (s, u, v) runs over the unit sphere in $\mathbb{R}^3$, and the real and imaginary coordinates in (3.1) are the image of that sphere under

$$L(s, u, v) = \frac{1}{2}(ds + cu, cv), \quad d = |\lambda_1 - \lambda_2|.$$

The image of the unit sphere under a linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$ is the same as the image of the unit ball: for a point with a preimage of norm below 1, add a vector in $\ker L$ until the preimage has norm 1. The image of the ball is a filled ellipse. Since

$$LL^T = \frac{1}{4} \begin{pmatrix} d^2 + c^2 & 0 \\ 0 & c^2 \end{pmatrix},$$

its semiaxes are

$$\frac{1}{2} \sqrt{|\lambda_1 - \lambda_2|^2 + c^2} \quad \text{and} \quad \frac{c}{2}.$$

The focal distance is $|\lambda_1 - \lambda_2|/2$, so the foci are λ_1, λ_2 . $\square$

Unitary Schur triangularization

Every complex square matrix B admits a unitary change of basis $U^*BU = R$ with R upper triangular; the diagonal entries of R are the eigenvalues of B . A standard proof chooses a unit eigenvector, puts it first in an orthonormal basis, and proceeds by induction on the lower-right compression. In dimension two, a diagonal unitary can rotate the single off-diagonal entry of R to a nonnegative real number. This is exactly the reduction used above.

3.3 Toeplitz–Hausdorff convexity

Theorem 3.4 (Toeplitz–Hausdorff). *The numerical range of every complex matrix is convex.*

Proof. Take $z_1, z_2 \in W(A)$, realized by unit vectors x_1, x_2 . Let $\mathcal{M} = \text{span}\{x_1, x_2\}$ and let $B = P_{\mathcal{M}}A|_{\mathcal{M}}$ be the compression of A to this subspace. If $\dim \mathcal{M} = 1$, then x_1 and x_2 differ only by a scalar phase, so $z_1 = z_2$ and there is nothing to prove. Otherwise $\dim \mathcal{M} = 2$. For every unit $x \in \mathcal{M}$,

$$x^* B x = x^* P_{\mathcal{M}} A x = x^* A x,$$

so $W(B) \subset W(A)$, while $z_1, z_2 \in W(B)$. By [theorem 3.3](#), $W(B)$ is convex and therefore contains the line segment joining z_1 and z_2 . The same segment lies in $W(A)$. $\square$

3.4 Support lines and Hermitian parts

For $\theta \in \mathbb{R}$, put

$$H_{\theta} = \text{Re}(e^{-i\theta} A) = \frac{e^{-i\theta} A + e^{i\theta} A^*}{2}.$$

Then

$$\max_{z \in W(A)} \text{Re}(e^{-i\theta} z) = \lambda_{\max}(H_{\theta}). \quad (3.2)$$

Indeed, the left side is

$$\max_{\|x\|=1} x^* H_{\theta} x,$$

which is the Rayleigh-quotient characterization of the largest Hermitian eigenvalue. Formula (3.2) both explains the geometry and gives a practical way to plot $W(A)$.

Proposition 3.5 (Exposed faces from the support pencil). *Fix $\theta \in \mathbb{R}$, and write*

$$h_{\theta} = \lambda_{\max}(H_{\theta}), \quad E_{\theta} = \ker(h_{\theta} I - H_{\theta}).$$

Let P_{θ} be the orthogonal projection onto E_{θ} , and define the Hermitian tangential compression

$$K_{\theta} = P_{\theta} \text{Im}(e^{-i\theta} A)|_{E_{\theta}}.$$

The exposed face of $W(A)$ with outward normal $e^{i\theta}$ is

$$\left\{ z \in W(A) : \text{Re}(e^{-i\theta} z) = h_{\theta} \right\} = e^{i\theta} (h_{\theta} + i[\lambda_{\min}(K_{\theta}), \lambda_{\max}(K_{\theta})]). \quad (3.3)$$

Proof. The support slack $h_{\theta} I - H_{\theta}$ is positive semidefinite. If a unit vector x generates a point on the exposed face, then

$$x^* (h_{\theta} I - H_{\theta}) x = 0.$$

By [theorem 1.4](#), this is equivalent to $x \in E_{\theta}$. Conversely, every unit vector in E_{θ} generates a point on the supporting line. For such a vector,

$$e^{-i\theta} x^* A x = h_{\theta} + i x^* \text{Im}(e^{-i\theta} A) x.$$

The Rayleigh quotients of K_{θ} fill the interval between its smallest and largest eigenvalues, which proves (3.3). $\square$

The planar boundary hides a spectral bundle

A simple top eigenvalue of H_θ produces one exposed point. A non-scalar K_θ produces a nontrivial flat edge. Thus the nullspace of the support slack records all vector states that generate the supported face; the planar outline shows only their scalar expectations.

3.5 The example that forces the constant 2

Let

$$N = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

For $x = (\cos t, e^{i\phi} \sin t)^T$,

$$x^* N x = e^{i\phi} \sin(2t).$$

As t and ϕ vary, this fills $\overline{\mathbb{D}}$. Hence

$$W(N) = \overline{\mathbb{D}}.$$

From Week 1, $\|N\| = 2$. Taking $p(z) = z$ gives

$$\|p(N)\| = 2 = 2 \max_{z \in W(N)} |p(z)|.$$

No universal constant smaller than 2 can work.

3.6 Continuity of the numerical range

Lemma 3.6 (Lipschitz inclusion). For $A, B \in \mathbb{M}_n(\mathbb{C})$,

$$W(B) \subset W(A) + \{z \in \mathbb{C} : |z| \leq \|B - A\|\}.$$

Proof. For every unit vector x ,

$$|x^*(B - A)x| \leq \|(B - A)x\|_2 \|x\|_2 \leq \|B - A\|.$$

□

This estimate will keep numerical ranges inside a fixed outer domain while matrices with simple spectrum approach an arbitrary matrix.

3.7 Exercises

Exercise 3.1 (Warm-up). Compute $W(A)$ when A is Hermitian. Then compute it when $A = \lambda I$.

Exercise 3.2 (Core: nilpotent disk). Fill in every step showing $W(N) = \overline{\mathbb{D}}$ for the matrix above.

Exercise 3.3 (Core: support function). Prove (3.2). Explain how a largest-eigenvalue routine can trace boundary points of $W(A)$.

Exercise 3.4 (Core: compression). If $\mathcal{M} \subset \mathbb{C}^n$ and $B = P_{\mathcal{M}} A|_{\mathcal{M}}$, prove $W(B) \subset W(A)$. Why is this the correct direction for the Toeplitz–Hausdorff proof?

Exercise 3.5 (Stretch: ellipse details). Verify the computation of LL^T after (3.1) and recover the two semiaxis lengths. Alternatively, obtain the same ellipse from its support function.

Chapter 4

Spectral sets and Crouzeix's problem

Learning goals

- State the notion of a C -spectral set.
- Explain why numerical ranges are natural domains for matrix functions.
- Separate the normal case from the nonnormal difficulty.
- Normalize the main inequality to a unit-bounded function.

Workshop plan

Opening: compare the spectrum and numerical range of the sharp nilpotent example. **Core:** spectral-set language and the constant question. **Guided proof:** normalize a polynomial to unit supremum. **Studio:** write the one-page problem brief, including the normal and nilpotent benchmark cases.

4.1 What a spectral set controls

Definition 4.1. Let $K \subset \mathbb{C}$ be compact and contain $\sigma(A)$. It is a C -spectral set for A if

$$\|r(A)\| \leq C \max_{z \in K} |r(z)|$$

for every rational function r with no pole on K .

Here rational matrix evaluation is concrete. Write $r = p/q$ with polynomials p, q . Having no pole on $K \supset \sigma(A)$ means $q(\lambda) \neq 0$ for every $\lambda \in \sigma(A)$; spectral mapping then makes $q(A)$ invertible, and

$$r(A) = p(A)q(A)^{-1}.$$

The value is independent of the chosen fraction and agrees with the holomorphic functional calculus.

The scalar maximum on the right is often accessible geometrically or numerically. The operator norm on the left measures the actual amplification of the matrix function. A spectral-set theorem transfers a scalar approximation guarantee to a matrix approximation guarantee.

Example 4.2 (Normal matrices). If A is normal, it is unitarily diagonalizable. Therefore

$$\|p(A)\| = \max_{\lambda \in \sigma(A)} |p(\lambda)| \leq \max_{z \in W(A)} |p(z)|.$$

Thus $W(A)$ gives constant 1 for normal matrices. The universal constant is forced above 1 only by nonnormal matrices.

4.2 The main theorem

Theorem 4.3 (Main theorem of the source manuscript). *For every $A \in \mathbb{M}_n(\mathbb{C})$ and every polynomial $p \in \mathbb{C}[z]$,*

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|. \quad (4.1)$$

Consequently, $W(A)$ is a 2-spectral set for A .

The factor 2 cannot be lowered, by the nilpotent example in Week 3. Historically, Crouzeix formulated the conjecture and proved foundational bounds [4, 5]; Crouzeix and Palencia obtained the universal constant $1 + \sqrt{2}$ [7]. The source manuscript's new mechanism is claimed to reach 2.

4.3 Normalization

Fix a bounded domain Ω containing $W(A)$, and set

$$m_\Omega = \max_{z \in \Omega} |p(z)|.$$

If $m_\Omega > 0$, let $f = p/m_\Omega$. Then

$$\max_{\Omega} |f| \leq 1, \quad p(A) = m_\Omega f(A).$$

Thus the core problem becomes:

$$\boxed{|f| \leq 1 \text{ on the domain} \implies \|f(A)\| \leq 2.}$$

This unit-ball normalization is what makes the Cayley transform in Week 9 have positive real part.

4.4 Why the numerical range is the right set

The spectrum is too small for nonnormal matrices: the nilpotent N has spectrum $\{0\}$ but norm 2. A large disk based only on $\|A\|$ is often too crude. The numerical range sits between:

$$\sigma(A) \subset W(A) \subset \{z : |z| \leq \|A\|\},$$

is compact and convex, changes continuously with A , and records directional information x^*Ax that the spectrum forgets.

4.5 Polynomial, holomorphic, and rational versions

The polynomial estimate is the core. Once it is known, approximation will give:

- the same bound for functions holomorphic near $W(A)$;
- the rational-function definition of a spectral set;
- a finite-dimensional-compression version for bounded operators on a Hilbert space.

These passages are deferred to Weeks 14–15 so that approximation does not obscure the new finite-dimensional argument.

Four layers of the proof

1. **Geometry:** put the numerical range in a smooth convex outer domain.
2. **Positivity:** use the double-layer measure and Cayley transforms to build a positive-real matrix function.
3. **Algebraic de-symmetrization:** cancel the adjoint-algebra defect and prove the norm bound 2.
4. **Approximation:** remove simple spectrum, algebra generation, and the outer domain.

4.6 Exercises

Exercise 4.1 (Warm-up). Show that if K is a C -spectral set for A , then $\alpha K + \beta$ is a C -spectral set for $\alpha A + \beta I$ when $\alpha \neq 0$.

Exercise 4.2 (Core: normal case). Prove carefully that a normal matrix satisfies (4.1) with constant 1.

Exercise 4.3 (Core: normalization). Explain separately what happens when $m_\Omega = 0$, and justify why normalizing by m_Ω loses no information.

Exercise 4.4 (Stretch: approximation error). Assume [theorem 4.3](#). If f is holomorphic near $W(A)$ and s is a polynomial, derive

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|.$$

List the additional fact needed to define $f(A)$.

Part II

Analytic and positive-function machinery

Chapter 5

Complex analysis and convex boundary geometry

Learning goals

- Read the contour integrals used in the functional calculus.
- Use Cauchy's formula and the maximum-modulus principle.
- Track boundary orientation, tangent, and outward normal.
- State the exact approximation theorem needed at the end.

Workshop plan

Opening: orient a circle and locate its outward normal. **Core:** holomorphic functions, Cauchy formula, matrix-valued integration. **Geometry lab:** supporting lines and $d\zeta = i\nu ds$. **Closing:** Runge approximation as a clearly marked classical input.

5.1 Holomorphic functions and contour integrals

A function $f : \Omega \rightarrow \mathbb{C}$, on an open set Ω , is *holomorphic* if its complex derivative exists at every point. Holomorphic functions are far more rigid than differentiable real functions: they have local power series, obey Cauchy's integral formula, and cannot attain a strict interior maximum of their modulus unless they are constant.

If Γ is a positively oriented simple closed C^1 curve and f is holomorphic on a neighborhood of the region it encloses, then

$$f(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta \quad (z \text{ inside } \Gamma). \quad (5.1)$$

Applying this entry by entry gives the matrix Cauchy formula. In particular, if $\sigma(B)$ lies inside Γ ,

$$\frac{1}{2\pi i} \int_{\Gamma} (\zeta I - B)^{-1} d\zeta = I. \quad (5.2)$$

Cauchy's theorem and formula

The standard proof first establishes that integrals of holomorphic functions around triangles vanish, deforms contours through a triangulation, and applies the result to $(f(\zeta) - f(z))/(\zeta - z)$. In this workshop, the scalar formula (5.1) is a classical input; every matrix use is a finite collection of scalar uses.

5.2 Maximum modulus

Theorem 5.1 (Maximum-modulus principle). *If f is holomorphic on a neighborhood of the closure of a bounded domain Ω , then*

$$\max_{z \in \bar{\Omega}} |f(z)| = \max_{\zeta \in \partial\Omega} |f(\zeta)|.$$

For the proof, the important consequence is that a boundary bound $|f| \leq 1$ controls the entire domain. We often state the bound on $\bar{\Omega}$ to avoid invoking the principle each time.

5.3 Smooth convex domains and normals

Let $\Omega \subset \mathbb{C} \cong \mathbb{R}^2$ be bounded and convex, with continuously differentiable boundary. At $\zeta \in \partial\Omega$, let $\nu(\zeta) \in \mathbb{C}$ be the outward unit normal and let ds denote arclength. When the boundary is traversed counterclockwise, the unit tangent is $i\nu$, so

$$d\zeta = i\nu(\zeta) ds, \quad d\bar{\zeta} = -i\overline{\nu(\zeta)} ds. \quad (5.3)$$

Convexity says the tangent line supports the whole domain. In complex notation,

$$\operatorname{Re}(\overline{\nu(\zeta)}(\zeta - z)) \geq 0 \quad (z \in \bar{\Omega}, \zeta \in \partial\Omega). \quad (5.4)$$

The expression is simply the outward normal component of the vector from z to ζ .

Example 5.2 (The disk). For $\Omega = \{z : |z| < R\}$, write $\zeta = Re^{it}$. Then $\nu = e^{it}$, $ds = R dt$, and

$$i\nu ds = iRe^{it} dt = d\zeta.$$

The support inequality becomes $\operatorname{Re}(R - e^{-it}z) \geq 0$, which follows from $|z| \leq R$.

Orientation signs

The identities for $d\zeta$ and $d\bar{\zeta}$ have opposite signs. The second sign is needed when the companion integral is adjointed in Week 8.

5.4 Uniform convergence and matrix evaluation

Suppose $f_j \rightarrow f$ uniformly on a contour Γ enclosing $\sigma(A)$. From the Cauchy functional calculus,

$$\|f_j(A) - f(A)\| \leq \frac{\operatorname{length}(\Gamma)}{2\pi} \max_{\zeta \in \Gamma} |f_j(\zeta) - f(\zeta)| \max_{\zeta \in \Gamma} \|(\zeta I - A)^{-1}\|.$$

Thus $f_j(A) \rightarrow f(A)$. This elementary estimate is the bridge from scalar approximation to matrix approximation.

Runge approximation in the needed form

Let $K \subset \mathbb{C}$ be compact and suppose $\mathbb{C} \setminus K$ is connected. If f is holomorphic on a neighborhood of K , then there are polynomials p_j converging uniformly to f on K .

We use this only when K is a compact convex neighborhood of a numerical range. Convexity makes the complement connected. A standard proof first approximates contour integrals by rational functions and then moves all poles to infinity when the complement is connected.

5.5 Distance, Hausdorff convergence, and outer domains

For a point z and set K ,

$$\text{dist}(z, K) = \inf_{w \in K} |z - w|.$$

For nonempty compact sets K, L , their Hausdorff distance is

$$d_H(K, L) = \max \left\{ \sup_{z \in K} \text{dist}(z, L), \sup_{w \in L} \text{dist}(w, K) \right\}.$$

The δ -parallel body

$$K + \delta \overline{\mathbb{D}} = \{k + \delta u : k \in K, |u| \leq 1\}$$

lies at Hausdorff distance δ from K . When K is convex, so is this parallel body. Week 13 will prove the regularity needed to use its interior as a double-layer domain.

5.6 Exercises

Exercise 5.1 (Warm-up: orientation). Verify (5.3) for an arclength parametrization $\gamma(s)$ of a counterclockwise C^1 curve.

Exercise 5.2 (Core: matrix Cauchy mass). Prove (5.2) first for a diagonalizable matrix and then explain why the holomorphic functional calculus gives it for every matrix.

Exercise 5.3 (Core: supporting line). Translate (5.4) into a dot-product inequality in $\mathbb{R}^2$, and prove it for an arbitrary convex set with a differentiable boundary.

Exercise 5.4 (Core: approximation transfer). Derive the displayed estimate for $\|f_j(A) - f(A)\|$ and identify every constant.

Exercise 5.5 (Stretch: Hausdorff maxima). Let K_j, K be nonempty compact sets contained in one compact set, with $d_H(K_j, K) \rightarrow 0$. Prove that for every continuous φ ,

$$\max_{K_j} \varphi \longrightarrow \max_K \varphi.$$

Chapter 6

Positive matrices, measures, and maps

Learning goals

- Define a positive matrix-valued measure and integrate against it.
- Recognize a positive and a unital positive map.
- Move positivity through integration.
- Understand the scalar Cayley transform as a positivity generator.

Workshop plan

Opening: turn a two-point scalar measure into a diagonal matrix measure. **Core:** positivity of measures and maps. **Guided proof:** matrix-measure Cauchy–Schwarz and contractivity. **Studio:** audit exactly where unitality and positivity are used.

6.1 From scalar measures to matrix-valued measures

A finite positive scalar measure μ assigns a nonnegative mass $\mu(E)$ to Borel sets and permits integration of continuous functions. A finite $\mathbb{M}_n(\mathbb{C})$ -valued measure M assigns a matrix $M(E)$ to each Borel set, countably additively entry by entry. It is *positive* if

$$M(E) \succeq 0 \quad \text{for every Borel set } E.$$

Equivalently, $u^*M(\cdot)u$ is a positive scalar measure for every $u \in \mathbb{C}^n$.

For a continuous scalar function φ , the matrix integral is defined entry by entry. Positivity gives

$$\varphi \geq 0 \implies \int \varphi dM \succeq 0, \tag{6.1}$$

because its quadratic form on u is the scalar integral against $u^*M(\cdot)u$.

Example 6.1 (A discrete positive matrix measure). Choose points $\zeta_1, \dots, \zeta_m$ and matrices $M_k \succeq 0$. Then

$$M = \sum_{k=1}^m M_k \delta_{\zeta_k}$$

is positive, and $\int \varphi dM = \sum_k \varphi(\zeta_k) M_k$.

6.2 Positive linear maps

For a compact space X , $C(X)$ denotes the complex-valued continuous functions on X , with pointwise addition and multiplication and the supremum norm $\|\varphi\|_\infty = \max_{x \in X} |\varphi(x)|$.

Definition 6.2. A complex-linear map $\Phi : C(X) \rightarrow \mathbb{M}_n(\mathbb{C})$, where X is compact, is *positive* if

$$\varphi \geq 0 \text{ pointwise} \implies \Phi(\varphi) \succeq 0.$$

It is *unital* if $\Phi(1) = I$.

If M is a positive matrix measure of total mass I , then

$$\Phi(\varphi) = \int_X \varphi dM$$

is unital and positive. Positive maps are automatically $*$ -preserving:

$$\Phi(\overline{\varphi}) = \Phi(\varphi)^*.$$

Indeed, a real-valued continuous function is a difference of two nonnegative functions, so its image is Hermitian; split a complex function into real and imaginary parts.

Lemma 6.3 (Contractivity of an integral positive map). *Let M be a positive $\mathbb{M}_n(\mathbb{C})$ -valued measure on X with $M(X) = I$, and define $\Phi(\varphi) = \int_X \varphi dM$. Then*

$$\|\Phi(\varphi)\| \leq \|\varphi\|_\infty.$$

Proof. For continuous vector-valued functions $F, G : X \rightarrow \mathbb{C}^n$, set

$$[F, G]_M = \int_X F(\zeta)^* dM(\zeta) G(\zeta).$$

Positivity of M makes this a positive semidefinite sesquilinear form, so the usual quadratic-polynomial proof gives Cauchy–Schwarz:

$$|[F, G]_M|^2 \leq [F, F]_M [G, G]_M.$$

For unit vectors x, y , choose $F(\zeta) = \overline{\varphi(\zeta)}x$ and $G(\zeta) = y$. Then

$$|x^* \Phi(\varphi) y| \leq \left(\int_X |\varphi|^2 x^* dM x \right)^{1/2} \left(\int_X y^* dM y \right)^{1/2} \leq \|\varphi\|_\infty.$$

Taking the supremum over unit x, y proves the claim. □

Common pitfall

A positive map is generally not multiplicative: $\Phi(f^2)$ need not equal $\Phi(f)^2$. The proof keeps $\Phi(f^m)$ as separate coefficients; it never makes this invalid replacement.

6.3 The scalar Cayley transform

For $|a| \leq 1$ and $w \in \mathbb{D}$, define

$$c_w(a) = \frac{1 + wa}{1 - \overline{w}a}.$$

The denominator is nonzero and

$$\operatorname{Re} c_w(a) = \frac{1 - |wa|^2}{|1 - \overline{w}a|^2} \geq 0. \quad (6.2)$$

Also,

$$c_w(a) = 1 + 2 \sum_{m=1}^{\infty} w^m a^m. \quad (6.3)$$

Thus the Cayley transform packages every power of a into one analytic function with positive real part.

Intuition

The unit disk is converted into the right half-plane. Disk bounds are hard nonlinear inequalities; positive real part is a linear matrix-order condition after applying a positive map. Retaining the whole parameter w preserves all power coefficients.

6.4 Exercises

Exercise 6.1 (Warm-up). Prove (6.2) by multiplying numerator and denominator by $1 - \overline{w}a$.

Exercise 6.2 (Core: positive integration). Prove (6.1) by testing on an arbitrary vector.

Exercise 6.3 (Core: star preservation). Give the complete proof that a positive complex-linear map $\Phi : C(X) \rightarrow \mathbb{M}_n(\mathbb{C})$ is $*$ -preserving.

Exercise 6.4 (Core: the factor two). Derive (6.3). Explain why the coefficient 2 will compensate for the $1/2$ in the definition of the double-layer map.

Exercise 6.5 (Stretch). Prove the Cauchy–Schwarz inequality for the form $[F, G]_M$ used in theorem 6.3, including the case in which $[G, G]_M = 0$.

Chapter 7

Positive-real functions and the Herglotz kernel

Learning goals

- Define a matrix-valued positive kernel.
- State scalar and matrix Herglotz representations.
- Prove that a positive-real analytic matrix function produces a positive kernel.
- Treat finite kernel sampling as an inequality-generating tool.

Workshop plan

Opening: scalar Cayley kernels. **Core:** positive kernels and scalar Herglotz. **Guided proof:** polarize scalar measures into a matrix measure. **Studio:** two-point and multi-point sampling.

7.1 Positive kernels

Definition 7.1. An $\mathbb{M}_n(\mathbb{C})$ -valued kernel $\mathcal{K}$ on $X \times X$ is *positive* if, for every finite choice of points $x_0, \dots, x_m \in X$ and vectors $\xi_0, \dots, \xi_m \in \mathbb{C}^n$,

$$\sum_{i,j=0}^m \xi_i^* \mathcal{K}(x_i, x_j) \xi_j \geq 0. \quad (7.1)$$

For scalar kernels this says every finite Gram matrix $[K(x_i, x_j)]$ is positive semidefinite. For matrix-valued kernels, the vectors ξ_i select both a point and a direction.

Example 7.2 (A factored kernel). If $V : X \rightarrow \mathbb{M}_{r,n}(\mathbb{C})$, then

$$\mathcal{K}(x, y) = V(x)^* V(y)$$

is positive, since the left side of (7.1) is $\left\| \sum_j V(x_j) \xi_j \right\|_2^2$. Integral mixtures of such kernels are positive as well.

7.2 The scalar Herglotz representation

Scalar Herglotz theorem

If $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic and $\operatorname{Re} f(w) \geq 0$, then there are a finite positive measure μ on $\mathbb{T}$ and a real number c such that

$$f(w) = ic + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} d\mu(\zeta). \quad (7.2)$$

They are unique.

Proof blueprint: the positive harmonic function $\operatorname{Re} f$ is the Poisson integral of a boundary measure obtained as a weak limit of its values on circles; a harmonic conjugate recovers f , up to the imaginary constant ic . Uniqueness follows from uniqueness of Fourier coefficients.

7.3 Assembling the matrix representation

Lemma 7.3 (Matrix Herglotz representation). *Let $F : \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ be analytic with $\operatorname{Re} F(w) \succeq 0$. Then there are a positive $\mathbb{M}_n(\mathbb{C})$ -valued measure M on $\mathbb{T}$ and a Hermitian matrix J_F such that*

$$F(w) = iJ_F + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} dM(\zeta). \quad (7.3)$$

Proof. For each $u \in \mathbb{C}^n$, the scalar function $u^* F(w) u$ has nonnegative real part. By the scalar Herglotz theorem, it has a positive measure μ_u and a real constant c_u . Uniqueness in the scalar theorem gives the measure identities

$$\mu_{\alpha u} = |\alpha|^2 \mu_u, \quad \mu_{u+v} + \mu_{u-v} = 2\mu_u + 2\mu_v.$$

Thus $u \mapsto \mu_u(E)$ is a nonnegative quadratic form for each Borel set E . Polarization defines

$$x^* M(E) y = \frac{1}{4} \sum_{k=0}^3 i^{-k} \mu_{x+ik y}(E). \quad (7.4)$$

Then $u^* M(E) u = \mu_u(E) \geq 0$, so M is positive. The imaginary constants polarize to

$$J_F = \frac{F(0) - F(0)^*}{2i},$$

which is Hermitian. Since all quadratic forms of the two sides agree, polarization assembles the scalar formulas into (7.3). $\square$

7.4 The Herglotz kernel

Lemma 7.4 (Matrix Herglotz kernel). *If $F : \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ is analytic and $\operatorname{Re} F(w) \succeq 0$, then*

$$\mathcal{L}_F(w, z) = \frac{F(w) + F(z)^*}{1 - w\bar{z}} \quad (7.5)$$

is a positive kernel on $\mathbb{D}$.

Proof. For $|\zeta| = 1$, direct algebra gives

$$\frac{1}{1 - w\bar{z}} \left(\frac{\zeta + w}{\zeta - w} + \frac{\overline{\zeta + z}}{\zeta - z} \right) = \frac{2}{(1 - w\bar{\zeta})(1 - \bar{z}\zeta)}. \quad (7.6)$$

The term iJ_F cancels when $F(w) + F(z)^*$ is formed. Substitution of (7.3) gives a factored integral. For finite samples w_i, ξ_i , the kernel quadratic form equals

$$2 \int_{\mathbb{T}} \left(\sum_i \frac{\xi_i}{1 - \bar{w}_i \zeta} \right)^* dM(\zeta) \left(\sum_j \frac{\xi_j}{1 - \bar{w}_j \zeta} \right) \geq 0.$$

This is precisely (7.1). □

A positive function becomes many inequalities

The pointwise condition $\operatorname{Re} F(w) \succeq 0$ is already useful. The Herglotz kernel is stronger operationally: every deliberate finite choice of points and vectors produces a scalar inequality. The new proof succeeds by choosing samples that annihilate what is unknown and retain what is useful.

7.5 Sampling as test design

The definition permits repeated points, zero vectors, and points chosen from spectral data. One should think of $\{(w_i, \xi_i)\}$ as a designed experiment:

1. split $F = F_{\text{known}} + F_{\text{nuisance}}$;
2. choose samples that make the nuisance quadratic form zero;
3. read a positive inequality from the known part.

Nothing says the kernel matrix of the known part alone is positive for all samples. Positivity is asserted only for the original kernel, and the cancellation works only for the special test data.

7.6 Exercises

Exercise 7.1 (Warm-up: factorization). Prove that every kernel $V(x)^* V(y)$ is positive.

Exercise 7.2 (Core: scalar identity). Verify (7.6), keeping track of $\bar{\zeta} = \zeta^{-1}$.

Exercise 7.3 (Core: polarization). Check (7.4) and show that its diagonal value is $\mu_u(E)$.

Exercise 7.4 (Core: two samples). Write the kernel inequality explicitly for points $0, w$ and vectors u, v . Identify the 2×2 block matrix whose quadratic form appears.

Exercise 7.5 (Stretch: direct Cayley atom). For $F(w) = (1 + wA)(1 - wA)^{-1}$ with a scalar $|A| \leq 1$, factor its Herglotz kernel directly and compare with (7.6).

Chapter 8

The classical double-layer positive map

Learning goals

- Construct the double-layer boundary measure.
- Prove its positivity from convex support lines.
- Recognize positivity as an exact support-slack test for numerical-range containment.
- Prove its total mass is $2I$.
- Derive the companion identity $2\Phi(h) = h(B) + g_h(B)^*$.
- Interpret the normalized measure as a boundary POVM and a symmetrized normal dilation.

Workshop plan

Opening: support lines become positive quadratic forms. **Core derivation:** resolvent density, positivity, and mass. **Contour lab:** companion transform with an interior contour. **Closing comparison:** what the classical symmetrized identity does and does not control.

8.1 Setup

Let $B \in \mathbb{M}_n(\mathbb{C})$, and let $\Omega \subset \mathbb{C}$ be a bounded convex domain with C^1 boundary such that

$$W(B) \subset \Omega.$$

For $\zeta \in \partial\Omega$, define

$$Z_\zeta = \zeta I - B, \quad \mathcal{R}_\zeta = Z_\zeta^{-1}.$$

The resolvent exists because $\sigma(B) \subset W(B) \subset \Omega$.

8.2 The boundary density is positive

Define the $\mathbb{M}_n(\mathbb{C})$ -valued boundary measure

$$d\mu_B(\zeta) = \frac{1}{2\pi} \left(\nu(\zeta) \mathcal{R}_\zeta + \overline{\nu(\zeta)} \mathcal{R}_\zeta^* \right) ds. \quad (8.1)$$

We show the matrix density in parentheses is positive.

For a unit vector x , the supporting-line inequality gives

$$2 \operatorname{Re} \left(\overline{\nu(\zeta)} (\zeta - x^* B x) \right) \geq 0.$$

Equivalently,

$$\nu(\zeta) Z_\zeta^* + \overline{\nu(\zeta)} Z_\zeta \succeq 0. \quad (8.2)$$

Congruence on the left by $\mathcal{R}_\zeta^*$ and on the right by $\mathcal{R}_\zeta$ turns the left side into

$$\nu(\zeta) \mathcal{R}_\zeta + \overline{\nu(\zeta)} \mathcal{R}_\zeta^*.$$

Thus $d\mu_B$ is a positive matrix measure.

Intuition

Convexity says every point $x^* B x$ lies behind the tangent line. The resolvent congruence lifts that scalar geometric fact into a positive matrix density. This is where numerical-range geometry enters the analytic machinery.

Proposition 8.1 (Exact support-slack test). *For this proposition only, do not assume the standing inclusion $W(B) \subset \Omega$. Let Ω be a bounded convex domain with C^1 boundary, and assume $\partial\Omega \cap \sigma(B) = \emptyset$. At $\zeta \in \partial\Omega$, set $u = \nu(\zeta)$ and*

$$h_\Omega(u) = \max_{z \in \Omega} \operatorname{Re}(\bar{u}z) = \operatorname{Re}(\bar{u}\zeta), \quad D_{\Omega,B}(u) = h_\Omega(u)I - \operatorname{Re}(\bar{u}B).$$

Then the density in (8.1) has the factorization

$$d\mu_B(\zeta) = \frac{1}{\pi} \mathcal{R}_\zeta^* D_{\Omega,B}(u) \mathcal{R}_\zeta ds. \quad (8.3)$$

Moreover,

$$\frac{d\mu_B}{ds}(\zeta) \succeq 0 \text{ for every } \zeta \in \partial\Omega \iff W(B) \subseteq \overline{\Omega}.$$

Proof. The identity

$$\nu Z_\zeta^* + \bar{\nu} Z_\zeta = 2D_{\Omega,B}(\nu)$$

and congruence by the invertible resolvent give (8.3). Thus positivity of the density is equivalent to

$$\operatorname{Re}(\bar{u}B) \preceq h_\Omega(u)I$$

for every boundary normal u . By (3.2), this says that every supporting half-plane of Ω contains $W(B)$. The intersection of those half-planes is $\overline{\Omega}$, proving the equivalence. $\square$

Remark 8.2. If $W(B)$ is compactly contained in Ω , then every $D_{\Omega,B}(u)$ is positive definite, uniformly in u . Moreover, since $\mathcal{R}_\zeta$ is invertible,

$$\operatorname{rank} \left(\frac{d\mu_B}{ds}(\zeta) \right) = \operatorname{rank} D_{\Omega,B}(u).$$

If $h_\Omega(u) = h_B(u)$, where

$$h_B(u) = \lambda_{\max}(\operatorname{Re}(\bar{u}B)),$$

then

$$\ker D_{\Omega,B}(u) = \ker (h_B(u)I - \operatorname{Re}(\bar{u}B)).$$

Thus the nullity records the dimension of the supporting-state space. The tangential compression from theorem 3.5 then determines the endpoints of the exposed face.

8.3 The mass is $2I$

Using $d\zeta = i\nu ds$ and the matrix Cauchy formula,

$$\frac{1}{2\pi} \int_{\partial\Omega} \nu(\zeta) \mathcal{R}_\zeta ds = \frac{1}{2\pi i} \int_{\partial\Omega} (\zeta I - B)^{-1} d\zeta = I.$$

Taking adjoints gives the same mass for the second layer. Hence

$$\int_{\partial\Omega} d\mu_B = 2I. \quad (8.4)$$

Define

$$\Phi(\varphi) = \frac{1}{2} \int_{\partial\Omega} \varphi(\zeta) d\mu_B(\zeta), \quad \varphi \in C(\partial\Omega). \quad (8.5)$$

By (8.4), $\Phi(1) = I$; positivity of the measure makes Φ positive. Thus Φ is unital, $*$ -preserving, and contractive.

Proposition 8.3 (Support–resolvent resolution of the identity). *Under the setup of this chapter,*

$$\int_{\partial\Omega} \mathcal{R}_\zeta^* D_{\Omega,B}(\nu(\zeta)) \mathcal{R}_\zeta ds = 2\pi I. \quad (8.6)$$

Equivalently, for every $x \in \mathbb{C}^n$,

$$\int_{\partial\Omega} \|D_{\Omega,B}(\nu(\zeta))^{1/2} \mathcal{R}_\zeta x\|_2^2 ds = 2\pi \|x\|_2^2. \quad (8.7)$$

Proof. Insert the factorization (8.3) into the mass identity (8.4). □

A support-weighted conservation law

Resolvent amplification is penalized by the support slack. The total slack-weighted boundary energy is exactly $2\pi I$, independently of the detailed shape of Ω .

8.4 The companion transform

For h holomorphic on a neighborhood of $\overline{\Omega}$, define

$$g_h(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{h(\zeta)}}{\zeta - z} d\zeta, \quad z \in \Omega. \quad (8.8)$$

The boundary data $\overline{h(\zeta)}$ are generally not holomorphic in ζ , but the integral defines a holomorphic function of the interior variable z .

Lemma 8.4 (Double-layer identity). *For h as above,*

$$2\Phi(h) = h(B) + g_h(B)^*. \quad (8.9)$$

Proof. The first layer of $2\Phi(h)$ is

$$\frac{1}{2\pi} \int_{\partial\Omega} h(\zeta) \nu(\zeta) (\zeta I - B)^{-1} ds = h(B)$$

by $d\zeta = i\nu ds$ and the holomorphic functional calculus.

For the second layer, do not assume that g_h extends across the boundary. Choose a contour $\Gamma \subset \Omega$ enclosing $\sigma(B)$. Cauchy's formula and finite-dimensional Fubini give

$$\begin{aligned} g_h(B) &= \frac{1}{2\pi i} \int_{\Gamma} g_h(z)(zI - B)^{-1} dz \\ &= \frac{1}{2\pi i} \int_{\partial\Omega} \overline{h(\zeta)}(\zeta I - B)^{-1} d\zeta. \end{aligned} \quad (8.10)$$

Taking adjoints and using $d\bar{\zeta} = -i\bar{\nu} ds$ identifies this with the second layer of $2\Phi(h)$. $\square$

8.5 The boundary POVM and a symmetrized normal dilation

The normalized measure

$$E_B(\mathcal{E}) = \frac{1}{2}\mu_B(\mathcal{E}), \quad \mathcal{E} \subseteq \partial\Omega \text{ Borel},$$

is a positive operator-valued measure with $E_B(\partial\Omega) = I$; in short, it is a POVM. The map Φ is integration against this POVM. Because its source $C(\partial\Omega)$ is commutative, a positive map from it is automatically completely positive.

Naimark dilation for the boundary POVM

There are a Hilbert space $\mathcal{K}$, an isometry $V : \mathbb{C}^n \rightarrow \mathcal{K}$, and a normal operator $\mathcal{N}$ on $\mathcal{K}$, with $\sigma(\mathcal{N}) \subseteq \partial\Omega$, such that

$$\Phi(\varphi) = V^* \varphi(\mathcal{N}) V \quad (\varphi \in C(\partial\Omega)).$$

This is the standard Naimark–Stinespring representation of a unital positive map on a commutative C^* -algebra. The dilation is not unique, and it is not needed for the norm proof.

In this representation, the double-layer identity becomes

$$h(B) + g_h(B)^* = 2V^*h(\mathcal{N})V. \quad (8.11)$$

Thus numerical-range containment naturally dilates the *symmetrized* analytic calculus. The obstruction is not the absence of a positive boundary model; it is the need to remove the coanalytic companion without treating it as an unrelated summand.

Common pitfall

The identity (8.9) is symmetrized. A bound on $\Phi(h)$ controls the sum $h(B) + g_h(B)^*$, not $h(B)$ alone. Estimating the companion as an unrelated summand throws away the algebraic coupling that the new proof exploits.

8.6 What is classical and what will be new

The positive boundary measure and companion transform are classical ingredients of the Delyon–Delyon and Crouzeix–Palencia framework [9, 7]. The new move is to apply Φ not just to f , but to the entire Cayley family

$$\frac{1 + wf}{1 - wf}, \quad w \in \mathbb{D}.$$

This records all companion terms $g_{f^m}(B)^*$ in one analytic resolvent completion.

8.7 Exercises

Exercise 8.1 (Warm-up: a scalar boundary point). For $\Omega = \mathbb{D}$, $B = 0$, and $\zeta = e^{it}$, compute $\nu(\zeta)$, $Z(\zeta)$, and the density $\bar{\nu}Z + \nu Z^*$ before reading off the mass of Φ .

Exercise 8.2 (Core: support to density). Write out the congruence that transforms (8.2) into the density in (8.1); verify the order of all noncommuting factors.

Exercise 8.3 (Core: two masses). Derive (8.4), showing separately the first and second layers.

Exercise 8.4 (Core: companion identity). Fill in the Fubini calculation leading to (8.10). Why is an interior contour Γ important?

Exercise 8.5 (Core: scalar disk). Take $n = 1$, $B = 0$, and $\Omega = \mathbb{D}$. Compute the density (8.1) and the map Φ explicitly.

Exercise 8.6 (Stretch: no multiplicativity). Find a simple positive unital map $\Phi : C(X) \rightarrow \mathbb{M}_2(\mathbb{C})$ and a function f for which $\Phi(f^2) \neq \Phi(f)^2$.

Part III

The positive-real completion mechanism

Chapter 9

The Cayley family creates a resolvent completion

Learning goals

- Build a matrix-valued Carathéodory function from the double-layer map.
- Expand it coefficient by coefficient.
- Locate the resolvent defect in $\text{alg}(T^*)$.
- State the abstract positive-real completion theorem.

Workshop plan

Opening: revisit the scalar Cayley series. **Core proof:** construct H , subtract the resolvent, and track the defect algebra. **Error clinic:** why $\Phi(f^m) \neq \Phi(f)^m$ is harmless. **Studio:** scalar and 2×2 completion examples.

9.1 Auxiliary hypotheses

For this week, assume:

- $B \in \mathbb{M}_n(\mathbb{C})$;
- Ω is a bounded convex C^1 domain with $W(B) \subset \Omega$;
- f is holomorphic on a neighborhood of $\overline{\Omega}$ and $\max_{\overline{\Omega}} |f| \leq 1$;
- $T = f(B)$ satisfies

$$\text{alg}(T) = \text{alg}(B). \tag{9.1}$$

The last condition is not automatic: a function can send different eigenvalues of B to the same value. Week 13 will enforce it by an arbitrarily small perturbation.

By spectral mapping,

$$\sigma(T) = f(\sigma(B)) \subset \overline{\mathbb{D}}.$$

9.2 Construction of the positive-real function

For $w \in \mathbb{D}$, define on the boundary

$$c_w(\zeta) = \frac{1 + wf(\zeta)}{1 - wf(\zeta)} \quad (9.2)$$

and put

$$H(w) = \Phi(c_w), \quad (9.3)$$

where Φ is the double-layer map from Week 8.

The map $w \mapsto c_w$ is analytic with values in $C(\partial\Omega)$, locally uniformly on $\mathbb{D}$. Since Φ is bounded and linear, H is analytic. Also

$$H(0) = \Phi(1) = I.$$

By (6.2),

$$\operatorname{Re} c_w(\zeta) = \frac{1 - |wf(\zeta)|^2}{|1 - wf(\zeta)|^2} \geq 0.$$

Because Φ is positive and $*$ -preserving,

$$\operatorname{Re} H(w) = \Phi(\operatorname{Re} c_w) \succeq 0. \quad (9.4)$$

Thus H is a normalized matrix-valued *Carathéodory function*: analytic on $\mathbb{D}$, equal to I at the origin, with positive semidefinite real part.

9.3 Coefficient ledger

The boundary expansion

$$c_w = 1 + 2 \sum_{m=1}^{\infty} w^m f^m$$

converges in the supremum norm on $\partial\Omega$, locally uniformly for $w \in \mathbb{D}$. Apply Φ term by term:

$$H(w) = I + 2 \sum_{m=1}^{\infty} w^m \Phi(f^m).$$

The double-layer identity, applied separately to each f^m , says

$$2\Phi(f^m) = f(B)^m + g_{f^m}(B)^* = T^m + g_{f^m}(B)^*.$$

Therefore

$$H(w) = I + \sum_{m=1}^{\infty} w^m (T^m + g_{f^m}(B)^*). \quad (9.5)$$

Why the coefficients are exactly right

The Cayley series contributes 2; the double-layer identity divides that coefficient between T^m and its companion. The known coefficients are T^m , so their sum is the resolvent:

$$I + \sum_{m \geq 1} w^m T^m = (I - wT)^{-1}.$$

9.4 The defect lies in the adjoint algebra

Because $\sigma(T) \subset \overline{\mathbb{D}}$ and $w \in \mathbb{D}$, the resolvent series converges in norm. Subtracting it from (9.5) gives

$$H(w) - (I - wT)^{-1} = \sum_{m=1}^{\infty} w^m g_{f^m}(B)^*. \quad (9.6)$$

Each g_{f^m} is holomorphic near $\sigma(B)$. By finite-dimensional holomorphic functional calculus,

$$g_{f^m}(B) \in \text{alg}(B).$$

Taking adjoints and using (9.1),

$$g_{f^m}(B)^* \in \text{alg}(B^*) = \text{alg}(T^*).$$

Every partial sum in (9.6) lies in $\text{alg}(T^*)$, and this finite-dimensional algebra is closed. Consequently,

$$H(w) - (I - wT)^{-1} \in \text{alg}(T^*) \quad (w \in \mathbb{D}). \quad (9.7)$$

Three different functions

c_w is a scalar boundary function of ζ ; $H(w)$ is a matrix function of w ; and $(I - wT)^{-1}$ is the known matrix resolvent. Keeping these roles separate prevents the false identity $H(w) = (I + wT)(I - wT)^{-1}$.

9.5 The abstract theorem now needed

Theorem 9.1 (Positive-real completion). *Let $T \in \mathbb{M}_n(\mathbb{C})$ have n distinct eigenvalues, all in $\overline{\mathbb{D}}$. Suppose $H : \mathbb{D} \rightarrow \mathbb{M}_n(\mathbb{C})$ is analytic and satisfies*

$$H(0) = I, \quad \text{Re } H(w) \succeq 0 \quad (w \in \mathbb{D}), \quad (9.8)$$

$$H(w) - (I - wT)^{-1} \in \text{alg}(T^*) \quad (w \in \mathbb{D}). \quad (9.9)$$

Then

$$\|T\| \leq 2.$$

This theorem is independent of numerical ranges and double-layer potentials. It is the new abstract tool. Weeks 10–12 prove it.

Corollary 9.2 (Auxiliary sharp bound). *Under the hypotheses of this chapter, if $T = f(B)$ has distinct eigenvalues, then $\|f(B)\| \leq 2$.*

Proof. The construction above verifies all hypotheses of [theorem 9.1](#). □

9.6 Why the full family matters

The single identity

$$2\Phi(f) = T + g_f(B)^*$$

contains one companion matrix. The full family (9.6) says much more:

- the known part is an entire resolvent;

- every unknown coefficient is adjoint-aligned;
- the unknowns form one analytic defect in a fixed finite algebra.

The completion theorem uses precisely this extra structure.

Remark 9.3 (Adjoint-algebraic unitary moment completion). Put $C_m = g_{f^m}(B)$. Comparing (9.5) with the matrix Herglotz representation gives a positive $\mathbb{M}_n(\mathbb{C})$ -valued measure M on $\mathbb{T}$. Because $H(0) = I$, the imaginary constant in the Herglotz representation vanishes and $M(\mathbb{T}) = I$. Moreover,

$$T^m + C_m^* = 2 \int_{\mathbb{T}} \bar{\zeta}^m dM(\zeta) \quad (m \geq 1).$$

Let F be a Naimark dilation of M , with isometry J , and define the unitary

$$U = \int_{\mathbb{T}} \bar{\zeta} dF(\zeta).$$

Then

$$T^m + C_m^* = 2J^*U^mJ \quad (m \geq 1). \quad (9.10)$$

Under the algebra-generation hypothesis, every C_m^* belongs to $\text{alg}(T^*)$. Thus the displayed positive-index sequence extends, by taking zeroth term $2I$ and negative terms as adjoints, to a positive-definite two-sided unitary moment sequence. The completion terms remain inside the opposite functional calculus.

For $\Omega = \mathbb{D}$ and $f(z) = z$, the boundary calculation gives $g_{z^m} = 0$ for every $m \geq 1$. In that case (9.10) reduces to the unitary 2-dilation identity $B^m = 2J^*U^mJ$. On a general convex domain, the companion C_m^* measures the failure of this disk Fourier splitting, while the proof shows that the failure remains algebraically constrained rather than arbitrary.

9.7 Exercises

Exercise 9.1 (Warm-up: Cayley coefficients). Expand $(1 + wa)/(1 - wa)$ through order w^3 and identify the coefficient of w^m for $m \geq 1$.

Exercise 9.2 (Core: positivity). Starting from (9.2), prove (9.4) without assuming that Φ is multiplicative.

Exercise 9.3 (Core: coefficient audit). Derive (9.5) from the Cayley and double-layer formulas, showing where both factors 2 go.

Exercise 9.4 (Core: algebra audit). Explain why Hermite rather than merely Lagrange interpolation may be needed to prove $g_{f^m}(B) \in \text{alg}(B)$ at this stage.

Exercise 9.5 (Core: scalar completion). When $n = 1$, write [theorem 9.1](#) in scalar language. What does the defect condition say, and why is the conclusion unsurprising?

Exercise 9.6 (Stretch: a wrong orientation). Let

$$T_M = \begin{pmatrix} 0 & M \\ 0 & 1/2 \end{pmatrix}, \quad H(w) \equiv I, \quad M > 0.$$

Show that

$$H(w) - (I - wT_M)^{-1} \in \text{alg}(T_M)$$

for all $w \in \mathbb{D}$, while $\|T_M\| \rightarrow \infty$ as $M \rightarrow \infty$. Thus $\text{alg}(T^*)$ cannot be replaced by $\text{alg}(T)$.

Chapter 10

Diagonalizing the structured nuisance term

Learning goals

- Convert the adjoint-algebra defect into a diagonal analytic correction.
- Preserve positivity using congruence.
- Understand the Gram matrix $G = S^*S$.
- Set up the special kernel samples.

Workshop plan

Opening: diagonalize one nonnormal 2×2 matrix and compute its Gram matrix. **Core:** pull the defect back by congruence and track the order ΘG . **Guided proof:** derive the half-eigenvalue and origin samples. **Studio:** reproduce the cancellation row by row, then complete the change-of-basis audit.

10.1 Diagonalization and its Gram matrix

Assume the hypotheses of [theorem 9.1](#). Since T has distinct eigenvalues,

$$T = S\Lambda S^{-1}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad G = S^*S. \quad (10.1)$$

The eigenvalues lie in $\overline{\mathbb{D}}$, and $G \succ 0$. The matrix G is the Gram matrix of the possibly nonorthogonal eigenbasis. It stores the nonnormal geometry that Λ alone cannot see.

By [theorem 2.3](#),

$$\text{alg}(T^*) = \left\{ (S^{-1})^* D S^* : D \text{ diagonal} \right\}. \quad (10.2)$$

10.2 The defect becomes diagonal

Let

$$\Delta(w) = H(w) - (I - wT)^{-1}.$$

For each w , membership in (10.2) gives a unique diagonal matrix $\Theta(w)$ such that

$$\Delta(w) = (S^{-1})^* \Theta(w) S^*.$$

Explicitly,

$$\Theta(w) = S^* \Delta(w) (S^*)^{-1}.$$

Therefore Θ is analytic. Since $H(0) = I$, $\Delta(0) = 0$, so

$$\Theta(0) = 0. \quad (10.3)$$

Write

$$\Theta(w) = \text{diag}(\theta_1(w), \dots, \theta_n(w)).$$

Now pull H back by congruence:

$$\tilde{H}(w) = S^* H(w) S.$$

The resolvent part transforms as

$$S^* (I - wT)^{-1} S = G(I - w\Lambda)^{-1},$$

while the defect becomes $\Theta(w)G$. Hence

$$\tilde{H}(w) = G(I - w\Lambda)^{-1} + \Theta(w)G. \quad (10.4)$$

Factor order matters

The nuisance term is $\Theta(w)G$, not $G\Theta(w)$. These matrices need not commute. The rowwise form ΘG is exactly what the sampling vectors can annihilate.

10.3 Positivity survives the change of coordinates

Similarity does not preserve positive real part, but congruence does:

$$\text{Re } \tilde{H}(w) = S^* (\text{Re } H(w)) S \succeq 0.$$

Thus the Herglotz kernel

$$\mathcal{L}_{\tilde{H}}(w, z) = \frac{\tilde{H}(w) + \tilde{H}(z)^*}{1 - w\bar{z}}$$

is positive by [theorem 7.4](#).

10.4 The two Cauchy-type matrices

Define Hermitian matrices P, Q, Y by

$$P_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j / 4}, \quad Q_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j / 2}, \quad Y = Q - P. \quad (10.5)$$

The denominators are nonzero because $|\bar{\lambda}_i \lambda_j| \leq 1$. Hermitian symmetry follows by conjugating the ij -entry.

At this point these definitions may look engineered. They will arise automatically from the two denominators in the sampled Herglotz kernel:

$$1 - w_i \bar{w}_j \quad \text{and} \quad 1 - w_i \lambda_j.$$

10.5 Choose the sample points and directions

Fix an arbitrary vector $u = (u_1, \dots, u_n)^T$. For $1 \leq i \leq n$, take

$$w_i = \frac{\overline{\lambda_i}}{2}, \quad \xi_i = u_i e_i. \quad (10.6)$$

Even when $|\lambda_i| = 1$, the sample lies strictly inside $\mathbb{D}$. Add one more sample at the origin:

$$w_0 = 0, \quad \xi_0 = v = -G^{-1}Pu. \quad (10.7)$$

The origin vector is allowed to depend on u . Kernel positivity holds for every finite collection of vectors, so this is legitimate.

10.6 Exact cancellation of the unknown function

Let $E(w) = \Theta(w)G$ be the nuisance part of (10.4). Because $\Theta(0) = 0$, all terms in the sampled kernel quadratic form involving $E(0)$ vanish. The remaining contribution of E and E^* is twice the real part of

$$\sum_{i=1}^n \overline{u_i} \theta_i(w_i) \left((Gv)_i + \sum_{j=1}^n \frac{G_{ij} u_j}{1 - w_i \overline{w_j}} \right). \quad (10.8)$$

Here is the index calculation:

$$e_i^* \Theta(w_i) G = \theta_i(w_i) e_i^* G,$$

so the diagonal correction selects row i of G . Moreover,

$$1 - w_i \overline{w_j} = 1 - \frac{\overline{\lambda_i} \lambda_j}{4}.$$

Therefore the sum in the large parentheses is $(Pu)_i$, and the whole parenthesis in (10.8) equals

$$(Gv + Pu)_i.$$

Our definition $v = -G^{-1}Pu$ makes this zero for every i . The entire unknown analytic correction disappears exactly.

Why one origin vector is enough

$$\begin{aligned} \Delta(w) &\in \text{alg}(T^*) \\ &\Downarrow \\ \Theta(w) &\text{ is diagonal} \\ &\Downarrow \\ &n \text{ unknown row directions} \\ &\Downarrow \\ Gv + Pu = 0 &\text{ is a system of exactly } n \text{ equations} \\ &\Downarrow \\ v = -G^{-1}Pu &\text{ cancels all rows simultaneously.} \end{aligned}$$

An arbitrary matrix-valued correction would have n^2 independent directions and would not be removable this way.

Cancellation is not vanishing

We have not proved $\Theta \equiv 0$. We have only shown that Θ 's contribution to one specially designed family of kernel quadratic forms is zero. That is all the proof needs.

10.7 Exercises

Exercise 10.1 (Warm-up: one row of a Gram matrix). For $S = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, compute $G = S^*S$, its inverse, and its two rows. Explain why ΘG scales rows whereas $G\Theta$ scales columns.

Exercise 10.2 (Core: transform the defect). Derive (10.4) from (10.1) and (10.2), showing every multiplication by S or S^* .

Exercise 10.3 (Core: analyticity and normalization). Prove that Θ is analytic and that (10.3) follows from $H(0) = I$.

Exercise 10.4 (Core: correction calculation). Starting from the definition of the Herglotz kernel, derive (10.8) without skipping the adjoint half.

Exercise 10.5 (Core: dimension count). Explain carefully why diagonal structure reduces the correction from n^2 scalar directions to n row directions.

Exercise 10.6 (Stretch: 2×2 symbolic check). Take arbitrary distinct $\lambda_1, \lambda_2 \in \overline{\mathbb{D}}$, a positive matrix $G = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$, and diagonal Θ . Write P , v , and the two cancellation equations explicitly.

Chapter 11

The surviving kernel inequality and two Gramians

Learning goals

- Compute every block of the sampled known kernel.
- Derive the anticommutator inequality for $Y = Q - P$.
- Balance the eigenbasis.
- Identify P and Q as weighted Gramians.

Workshop plan

Opening: verify the scalar partial fraction that creates P and Q . **Core:** assemble the three known kernel blocks. **Guided proof:** substitute the compensating vector and balance the basis. **Studio:** trace the worked 2×2 audit and then repeat it with new eigenvalues.

11.1 The sample–sample block

The known part of $\tilde{H}$ is

$$F(w) = G(I - w\Lambda)^{-1}.$$

At sample points $w_i = \bar{\lambda}_i/2$, the ij -entry of its Herglotz kernel is

$$\begin{aligned} \frac{F(w_i)_{ij} + F(w_j)_{ij}^*}{1 - w_i \bar{w}_j} &= \frac{2G_{ij}}{(1 - \bar{\lambda}_i \lambda_j/2)(1 - \bar{\lambda}_i \lambda_j/4)} \\ &= (4Q - 2P)_{ij}. \end{aligned} \tag{11.1}$$

The last equality is the scalar partial-fraction identity

$$\frac{2}{(1 - x/2)(1 - x/4)} = \frac{4}{1 - x/2} - \frac{2}{1 - x/4}.$$

11.2 The origin blocks

For a fixed eigenvalue sample, the literal sample–origin block is

$$\mathcal{L}_F(w_i, 0) = F(w_i) + G.$$

After applying the prescribed vector $\xi_i = u_i e_i$, its selected row is

$$e_i^*(F(w_i) + G) = e_i^*(Q + G).$$

Thus, after the n selected rows are assembled in the quadratic form, the *effective* cross block is

$$G + Q. \quad (11.2)$$

At the origin the literal and effective block is

$$2G. \quad (11.3)$$

After the exact cancellation in Week 10, positivity of the full Herglotz kernel therefore yields, for the particular $v = -G^{-1}Pu$,

$$0 \leq \begin{pmatrix} u \\ v \end{pmatrix}^* \begin{pmatrix} 4Q - 2P & G + Q \\ G + Q & 2G \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}. \quad (11.4)$$

Common pitfall

Equation (11.4) is asserted on the graph $v = -G^{-1}Pu$. The displayed known-part block matrix is not claimed to be positive semidefinite for arbitrary independent u, v ; the unknown part was canceled only for the special graph.

11.3 Substitute the compensator

Insert $v = -G^{-1}Pu$ into (11.4). Since P , Q , and G are Hermitian, the result is

$$0 \leq u^* [(4Q - 2P) - (G + Q)G^{-1}P - PG^{-1}(G + Q) + 2PG^{-1}P]u. \quad (11.5)$$

Use $Q = P + Y$. The pure P -terms cancel:

$$\begin{aligned} (4Q - 2P) - (G + Q)G^{-1}P - PG^{-1}(G + Q) + 2PG^{-1}P \\ = 4Y - YG^{-1}P - PG^{-1}Y. \end{aligned}$$

Since u was arbitrary,

$$4Y - YG^{-1}P - PG^{-1}Y \succeq 0. \quad (11.6)$$

This is the entire payload of the kernel sampling.

11.4 A fully worked 2×2 audit

The following small case makes the cancellation visible without hiding it in indices. Take

$$T = \begin{pmatrix} 0 & 1 \\ 0 & 1/2 \end{pmatrix} = S\Lambda S^{-1}, \quad S = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad \Lambda = \text{diag}(0, 1/2).$$

Then

$$G = S^*S = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \quad G^{-1} = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}.$$

The definitions (10.5) give

$$P = \begin{pmatrix} 1 & 2 \\ 2 & 16/3 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 2 \\ 2 & 40/7 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 0 & 8/21 \end{pmatrix}.$$

For $u = (u_1, u_2)^T$,

$$G^{-1}P = \begin{pmatrix} 1 & -2/3 \\ 0 & 4/3 \end{pmatrix}, \quad v = -G^{-1}Pu = \begin{pmatrix} -u_1 + \frac{2}{3}u_2 \\ -\frac{4}{3}u_2 \end{pmatrix}.$$

Consequently $Gv + Pu = 0$. Whatever the two analytic diagonal entries of Θ may be, their sampled contribution

$$2 \operatorname{Re} \sum_{i=1}^2 \bar{u}_i \theta_i(w_i) (Gv + Pu)_i$$

is exactly zero. The known remainder is positive because

$$4Y - YG^{-1}P - PG^{-1}Y = \begin{pmatrix} 0 & 0 \\ 0 & 32/63 \end{pmatrix} \succeq 0.$$

This calculation is an audit of the completion mechanism: it assumes the positive-real completion hypotheses, but it does not need to know the values of the nuisance functions θ_i .

11.5 Balance the eigenbasis

Define

$$\begin{aligned} \tilde{T} &= G^{1/2} \Lambda G^{-1/2}, & \hat{P} &= G^{-1/2} P G^{-1/2}, \\ \hat{Q} &= G^{-1/2} Q G^{-1/2}, & \hat{Y} &= \hat{Q} - \hat{P}. \end{aligned} \tag{11.7}$$

This is not an arbitrary normalization. The polar decomposition of S is

$$S = U G^{1/2}$$

with U unitary, and therefore

$$T = U \tilde{T} U^*. \tag{11.8}$$

Thus T and $\tilde{T}$ have the same operator norm.

Congruence of (11.6) by $G^{-1/2}$ gives

$$4\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} \succeq 0. \tag{11.9}$$

11.6 The two weighted Gramians

Expand the scalar denominators in (10.5) as geometric series. Entrywise,

$$P = \sum_{k=0}^{\infty} 4^{-k} \Lambda^{*k} G \Lambda^k, \quad Q = \sum_{k=0}^{\infty} 2^{-k} \Lambda^{*k} G \Lambda^k.$$

After congruence,

$$\hat{P} = \sum_{k=0}^{\infty} 4^{-k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Q} = \sum_{k=0}^{\infty} 2^{-k} \tilde{T}^{*k} \tilde{T}^k. \tag{11.10}$$

These are weighted observability Gramians: on a vector x ,

$$x^* \hat{P} x = \sum_{k \geq 0} 4^{-k} \|\tilde{T}^k x\|_2^2.$$

Even if some $|\lambda_j| = 1$, the series converge in norm. Indeed,

$$\tilde{T}^k = G^{1/2} \Lambda^k G^{-1/2},$$

so $\{\tilde{T}^k\}$ is uniformly bounded; the scalar weights are summable.

Subtracting the two series gives

$$\hat{Y} = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0. \quad (11.11)$$

Also, the $k = 0$ term in $\hat{P}$ gives

$$\hat{P} \succeq I. \quad (11.12)$$

What the two scales accomplish

The 1/2-weighted Gramian $\hat{Q}$ sees more of the future orbit than the 1/4-weighted Gramian $\hat{P}$. Their difference $\hat{Y}$ is a positive detector: if it sees zero energy, then already $\tilde{T}x = 0$. The narrower Gramian $\hat{P}$ also obeys the Stein recursion that will control $\|\tilde{T}\|$.

11.7 Exercises

Exercise 11.1 (Warm-up: the two denominators). With $x = 1/2$, evaluate $2/[(1 - x/2)(1 - x/4)]$ directly and from $4/(1 - x/2) - 2/(1 - x/4)$.

Exercise 11.2 (Core: top-left block). Derive (11.1) entry by entry. Check both the resolvent denominator and the Herglotz denominator.

Exercise 11.3 (Core: graph substitution). Starting with (11.4), derive (11.5) and then (11.6). Preserve the order of all factors.

Exercise 11.4 (Core: balance). Prove (11.8) from the polar decomposition of S .

Exercise 11.5 (Core: Gramian expansion). Prove both identities in (11.10), starting from the entrywise formulas for P, Q .

Exercise 11.6 (Stretch: kernel of the detector). Use (11.11) to prove

$$\ker \hat{Y} \subseteq \ker \tilde{T}.$$

Then show $\ker \hat{Y} \subseteq \ker(\hat{P} - I)$.

Chapter 12

Gramian rigidity, the Stein identity, and the constant 2

Learning goals

- Convert an anticommutator inequality into a spectral bound.
- Use a positive detector even when it is singular.
- Derive and exploit a Stein identity.
- Explain conceptually why the sampling radius $1/2$ is optimal within the one-parameter family studied below.
- Interpret the two factors $\sqrt{2}$ as a controlled similarity and a transformed norm.

Workshop plan

Opening: test the anticommutator inequality on eigenvectors of P . **Core:** detector rigidity and its singular case. **Guided proof:** shift the Gramian series to obtain the Stein identity. **Oral studio:** reconstruct the completion theorem from the seven-item checklist.

12.1 A reusable rigidity lemma

Lemma 12.1 (Detector–anticommutator rigidity). *Let $P = P^*$ and $X = X^*$ satisfy*

$$P \succeq I, \quad X \succeq 0, \quad cX - XP - PX \succeq 0, \quad c \geq 2. \quad (12.1)$$

Assume also the detectability condition

$$\ker X \subseteq \ker(P - I). \quad (12.2)$$

Then

$$P \preceq \frac{c}{2}I.$$

Proof. Choose a unit eigenvector e of the Hermitian matrix P , with $Pe = \alpha e$. Testing the anticommutator inequality on e gives

$$0 \leq (c - 2\alpha)e^*Xe.$$

If $\alpha > c/2$, the coefficient is negative. Since $X \succeq 0$, we must have $e^*Xe = 0$. By [theorem 1.4](#), $Xe = 0$, and detectability gives $Pe = e$. This contradicts $\alpha > c/2 \geq 1$. Hence every eigenvalue of P is at most $c/2$. $\square$

Common pitfall

The anticommutator inequality alone is not enough when X is singular. The detector condition (12.2) supplies the missing rigidity on $\ker X$.

12.2 Apply rigidity to the two Gramians

Take

$$P = \hat{P}, \quad X = \hat{Y}, \quad c = 4.$$

The lower bound $P \succeq I$ is (11.12); positivity $X \succeq 0$ is (11.11); and the anticommutator inequality is (11.9).

Detectability follows from the series. If $\hat{Y}e = 0$, then $e^*\hat{Y}e = 0$, and (11.11) gives

$$0 = \sum_{k=1}^{\infty} (2^{-k} - 4^{-k}) \|\tilde{T}^k e\|_2^2.$$

Every coefficient is positive, so $\tilde{T}e = 0$. The series for $\hat{P}$ then yields $\hat{P}e = e$. Thus (12.2) holds, and [theorem 12.1](#) gives

$$I \preceq \hat{P} \preceq 2I. \quad (12.3)$$

12.3 The Stein identity

Shift the first Gramian series:

$$\begin{aligned} \hat{P} - I &= \sum_{k=1}^{\infty} 4^{-k} \tilde{T}^{*k} \tilde{T}^k \\ &= \frac{1}{4} \tilde{T}^* \left(\sum_{j=0}^{\infty} 4^{-j} \tilde{T}^{*j} \tilde{T}^j \right) \tilde{T}. \end{aligned}$$

Hence

$$\hat{P} = I + \frac{1}{4} \tilde{T}^* \hat{P} \tilde{T}. \quad (12.4)$$

Using (12.3),

$$\tilde{T}^* \tilde{T} \preceq \tilde{T}^* \hat{P} \tilde{T} = 4(\hat{P} - I) \preceq 4I.$$

Therefore

$$\|\tilde{T}\| \leq 2.$$

Finally $T = U\tilde{T}U^*$ by (11.8), so

$$\boxed{\|T\| \leq 2.}$$

This completes the proof of [theorem 9.1](#).

The sharp chain

$$\begin{aligned} \text{positive kernel} &\implies 4\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} \succeq 0, \\ \hat{Y} \text{ detects } \tilde{T} &\implies I \preceq \hat{P} \preceq 2I, \\ \hat{P} = I + \tfrac{1}{4}\tilde{T}^*\hat{P}\tilde{T} &\implies \tilde{T}^*\tilde{T} \preceq 4I. \end{aligned}$$

The constant 2 is the square root of the final coefficient 4.

Remark 12.2 (Equality cannot occur at the auxiliary stage). If $\|\tilde{T}\| = 2$, choose a unit norm-attaining vector e . Equality throughout the Stein chain forces

$$\hat{P}e = 2e, \quad \hat{P}\tilde{T}e = \tilde{T}e.$$

The series for $\hat{P} - I$ then gives $\tilde{T}^2e = 0$, while $\|\tilde{T}e\| = 2$. This is a nontrivial Jordan chain at zero. But the completion theorem assumes simple spectrum, hence diagonalizability, and a diagonalizable matrix satisfies $\ker \tilde{T}^2 = \ker \tilde{T}$. Therefore the auxiliary estimate is actually strict:

$$\|T\| < 2.$$

Accordingly, any equality obtained after the approximation steps must emerge as diagonalizability is lost in the limiting argument.

12.4 Why one half is the distinguished radius

The source proof samples at half the conjugate eigenvalues. A one-parameter version explains this choice. Fix $0 < r < 1$ and instead sample

$$w_i = r\bar{\lambda}_i.$$

Define

$$(P_r)_{ij} = \frac{G_{ij}}{1 - r^2\bar{\lambda}_i\lambda_j}, \quad (Q_r)_{ij} = \frac{G_{ij}}{1 - r\bar{\lambda}_i\lambda_j}, \quad Y_r = Q_r - P_r. \quad (12.5)$$

The same origin choice $v = -G^{-1}P_ru$ cancels the diagonal correction. The sample-sample block uses

$$\frac{2}{(1-rx)(1-r^2x)} = \frac{2}{1-r} \frac{1}{1-rx} - \frac{2r}{1-r} \frac{1}{1-r^2x}. \quad (12.6)$$

Thus the three assembled known blocks are

$$\frac{2}{1-r}Q_r - \frac{2r}{1-r}P_r, \quad G + Q_r, \quad 2G,$$

in the sample-sample, cross, and origin positions, respectively. Substitute $v = -G^{-1}P_ru$ into their quadratic form and use $Q_r = P_r + Y_r$. The terms containing only P_r cancel, leaving

$$\frac{2}{1-r}Y_r - Y_rG^{-1}P_r - P_rG^{-1}Y_r \succeq 0. \quad (12.7)$$

Define the balanced matrices

$$\hat{P}_r = G^{-1/2}P_rG^{-1/2}, \quad \hat{Y}_r = G^{-1/2}Y_rG^{-1/2}.$$

Congruence of (12.7) gives

$$\frac{2}{1-r}\hat{Y}_r - \hat{Y}_r\hat{P}_r - \hat{P}_r\hat{Y}_r \succeq 0, \quad (12.8)$$

where

$$\hat{P}_r = \sum_{k=0}^{\infty} r^{2k} \tilde{T}^{*k} \tilde{T}^k, \quad (12.9)$$

$$\hat{Y}_r = \sum_{k=1}^{\infty} (r^k - r^{2k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0. \quad (12.10)$$

The same detector argument applies with $c = 2/(1-r)$, giving

$$I \preceq \hat{P}_r \preceq \frac{1}{1-r}I.$$

The Stein identity now reads

$$\hat{P}_r = I + r^2 \tilde{T}^* \hat{P}_r \tilde{T}.$$

Therefore

$$r^2 \tilde{T}^* \tilde{T} \preceq \hat{P}_r - I \preceq \frac{r}{1-r}I,$$

and

$$\|\tilde{T}\| \leq \frac{1}{\sqrt{r(1-r)}}. \quad (12.11)$$

The product $r(1-r)$ is uniquely maximized on $(0,1)$ at $r = 1/2$. Substituting that value into (12.11) gives the best bound produced by this family:

$$\|\tilde{T}\| \leq 2.$$

The canonical orbit Gramian and a controlled similarity

The raw matrix P_r depends on the chosen eigenbasis, but the balanced Gramian has a canonical, unitarily covariant form. Since $T = U\tilde{T}U^*$, define

$$\mathcal{P}_r(T) = \sum_{k=0}^{\infty} r^{2k} T^{*k} T^k = U \hat{P}_r U^*. \quad (12.12)$$

It satisfies

$$I \preceq \mathcal{P}_r(T) \preceq \frac{1}{1-r}I, \quad \mathcal{P}_r(T) = I + r^2 T^* \mathcal{P}_r(T) T. \quad (12.13)$$

Put $X_r = \mathcal{P}_r(T)^{1/2}$. The second identity gives

$$(X_r T X_r^{-1})^* (X_r T X_r^{-1}) = \frac{1}{r^2} (I - \mathcal{P}_r(T)^{-1}) \preceq \frac{1}{r}I,$$

while the first gives

$$\kappa(X_r) = \|X_r\| \|X_r^{-1}\| \leq \frac{1}{\sqrt{1-r}}.$$

Thus T is similar, through a quantitatively controlled positive metric, to an operator of norm at most $r^{-1/2}$. Multiplying the similarity bound and the transformed norm recovers

$$\|T\| \leq \frac{1}{\sqrt{1-r}} \frac{1}{\sqrt{r}}.$$

At $r = 1/2$, both factors equal $\sqrt{2}$. The constant 2 is therefore the balanced product of a metric-conditioning cost and a norm-in-that-metric cost.

Workshop extension

The parameter calculation in this section is a pedagogical extension of the source's half-eigenvalue sampling. It shows that $1/2$ is not a lucky guess within this one-parameter family: it optimally balances how strongly the detector sees the orbit against how strongly the Stein recursion amplifies the norm.

12.5 Why the same-side algebra is insufficient

Return to the matrices in [exercise 9.6](#):

$$T_M = \begin{pmatrix} 0 & M \\ 0 & 1/2 \end{pmatrix}, \quad H(w) \equiv I, \quad M > 0.$$

They have distinct eigenvalues in $\mathbb{D}$, and H is normalized positive-real. Moreover,

$$H(w) - (I - wT_M)^{-1} = - \sum_{k=1}^{\infty} w^k T_M^k \in \text{alg}(T_M).$$

Yet $\|T_M\| \geq M$. Thus a same-side algebra constraint gives no uniform bound. The adjoint orientation in $\text{alg}(T^*)$ is what becomes row-diagonal after the positivity-preserving congruence.

12.6 The reusable new-tool template

The proof suggests the following five-step design pattern.

Structured kernel cancellation–Gramian rigidity

1. **Find a positive kernel** generated by a normalized positive-real analytic function.
2. **Identify an adjoint-aligned nuisance module** that has fewer effective directions than an arbitrary matrix correction.
3. **Build an annihilating sample**—often by adding an anchor point whose vector solves a linear system.
4. **Recognize the surviving terms as ordered weighted Gramians**; their difference should be positive and detect the dynamics.
5. **Use a Stein recursion** to turn the Gramian order bound into a sharp operator-norm estimate.

The method is credible only when every stage can be verified without estimating the nuisance term.

12.7 Core-proof oral checklist

A participant is ready to proceed when they can answer, without notes:

1. Why does simple spectrum make the defect diagonal after the chosen congruence?
2. Why is $\Theta(0) = 0$ essential?

3. Where do the denominators $1 - x/4$ and $1 - x/2$ come from?
4. Why does $v = -G^{-1}Pu$ cancel every $\theta_i(w_i)$?
5. Why is $\hat{Y} \succeq 0$?
6. Why does $\hat{Y}e = 0$ imply $\hat{P}e = e$?
7. Which identity finally turns $2I - \hat{P} \succeq 0$ into $\|T\| \leq 2$?

12.8 Exercises

Exercise 12.1 (Warm-up: a scalar detector). Let $P = \alpha \geq 1$, $X = \beta \geq 0$, and $cX - XP - PX \geq 0$. Determine what follows when $\beta > 0$ and what additional information is needed when $\beta = 0$.

Exercise 12.2 (Core: rigidity). Prove [theorem 12.1](#) from scratch, including the use of [theorem 1.4](#).

Exercise 12.3 (Core: Stein shift). Derive [\(12.4\)](#) directly from the series and use it to reproduce the last operator inequality.

Exercise 12.4 (Core: general radius). Verify [\(12.6\)](#), derive [\(12.8\)](#), and optimize [\(12.11\)](#).

Exercise 12.5 (Core: wrong algebra). Complete [exercise 9.6](#) and explain why commutativity of $\text{alg}(T)$ does not rescue the theorem.

Exercise 12.6 (Stretch: equality clues). Without consulting [theorem 12.2](#), reconstruct the equality analysis as follows. Assume $\|\tilde{T}\| = 2$ and choose a unit vector e with $\|\tilde{T}e\| = 2$. Trace equality through the final chain to show

$$\hat{P}e = 2e, \quad \hat{P}\tilde{T}e = \tilde{T}e.$$

Deduce $\tilde{T}^2e = 0$, and explain why this contradicts the simple-spectrum hypothesis. Then compare with the limiting nilpotent restriction in [theorem 14.4](#). Finally, examine the diagonalizable matrices

$$T_\varepsilon = \begin{bmatrix} \varepsilon & 2 \\ 0 & -\varepsilon \end{bmatrix}, \quad \varepsilon \neq 0,$$

and explain how $T_\varepsilon \rightarrow \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ illustrates the loss of diagonalizability at an extremal limit.

Part IV

Removing hypotheses and completing the theorem

Chapter 13

Perturbation and smooth convex outer approximation

Learning goals

- Approximate arbitrary matrices by simple-spectrum matrices.
- Force algebra generation by perturbing the scalar function.
- Construct C^1 convex outer domains.
- Interpret parallel-body smoothing as a uniform support-slack regularization and derive an averaged resolvent bound.
- Keep a precise ledger of which limit removes which hypothesis.

Workshop plan

Opening: list every auxiliary hypothesis still present. **Core I:** simple spectrum and the separating perturbation f_η . **Core II:** parallel bodies and C^1 boundary. **Studio:** write an explicit three-limit ledger.

13.1 Simple-spectrum matrices are dense

Lemma 13.1 (Density of simple spectrum). *Matrices with n distinct eigenvalues are dense in $\mathbb{M}_n(\mathbb{C})$.*

Proof. The discriminant of a monic polynomial vanishes exactly when the polynomial has a repeated root. Applied to the characteristic polynomial, it is a polynomial in the entries of the matrix. It is not the zero polynomial: at $\text{diag}(1, 2, \dots, n)$ it is nonzero. The zero set of a nonzero complex polynomial has empty interior. Therefore every neighborhood in $\mathbb{M}_n(\mathbb{C})$ contains a matrix whose characteristic discriminant is nonzero, hence whose eigenvalues are distinct. $\square$

Why a polynomial zero set has no interior

For one complex variable, a nonzero polynomial has only finitely many zeros. For several variables, induct on the number of variables: write the polynomial as a polynomial in the last

variable whose coefficients are polynomials in the others. If it vanished on an open polydisk, the one-variable statement would force every coefficient to vanish on an open set; induction would make every coefficient the zero polynomial. This contradicts the assumption. Applied to the real and imaginary matrix entries, this is the density fact used above.

Combine this with [theorem 3.6](#). If $W(A)$ is compactly contained in an open domain Ω , set

$$\delta_\Omega = \text{dist}(W(A), \mathbb{C} \setminus \Omega) > 0.$$

Whenever $\|B - A\| < \delta_\Omega$,

$$W(B) \subset W(A) + \|B - A\| \overline{\mathbb{D}} \subset \Omega.$$

Thus we may choose simple-spectrum $B_k \rightarrow A$ while keeping all their numerical ranges in the same domain.

13.2 Why $f(B)$ may not generate $\text{alg}(B)$

Let B have distinct eigenvalues $\beta_1, \dots, \beta_n$. If $f(\beta_i) = f(\beta_j)$ for some $i \neq j$, then $T = f(B)$ has repeated eigenvalues and cannot recover the two corresponding spectral coordinates. Both hypotheses needed by the completion theorem can fail at once:

$$T \text{ may not have simple spectrum,} \quad \text{alg}(T) \text{ may be smaller than } \text{alg}(B).$$

The remedy separates the values of f without breaking the unit bound.

13.3 The algebra-generating perturbation

Fix a bounded convex domain Ω , and suppose $\max_{\overline{\Omega}} |f| \leq 1$. Put

$$R_\Omega = \max_{z \in \overline{\Omega}} |z|.$$

For $\eta \in \mathbb{C}$, define

$$f_\eta(z) = \frac{f(z) + \eta z}{1 + |\eta|R_\Omega}. \quad (13.1)$$

For each fixed η , this is holomorphic in z , and

$$|f_\eta(z)| \leq \frac{1 + |\eta|R_\Omega}{1 + |\eta|R_\Omega} = 1 \quad (z \in \overline{\Omega}).$$

The appearance of $|\eta|$ affects dependence on the parameter η , but not holomorphicity in the variable z .

For $i \neq j$, the collision equation

$$f(\beta_i) + \eta\beta_i = f(\beta_j) + \eta\beta_j$$

excludes at most one value of η , because $\beta_i \neq \beta_j$. There are finitely many pairs, hence finitely many exceptional parameters. We can choose nonexceptional $\eta_\ell \rightarrow 0$.

Set

$$T_\ell = f_{\eta_\ell}(B).$$

Its eigenvalues $f_{\eta_\ell}(\beta_i)$ are distinct, so T_ℓ has simple spectrum. Also $T_\ell \in \text{alg}(B)$. Conversely, Lagrange interpolation gives a polynomial q_ℓ such that

$$q_\ell(f_{\eta_\ell}(\beta_i)) = \beta_i \quad (1 \leq i \leq n).$$

Because B is diagonalizable,

$$q_\ell(T_\ell) = B.$$

Therefore

$$\text{alg}(T_\ell) = \text{alg}(B). \quad (13.2)$$

Common pitfall

Adding ηz without the denominator in (13.1) separates eigenvalue images but may violate $|f_\eta| \leq 1$. The denominator is what keeps the Cayley transform positive-real.

13.4 Fixed-domain estimate

Proposition 13.2 (Estimate on one smooth outer domain). *Let $A \in \mathbb{M}_n(\mathbb{C})$, let p be a polynomial, and let Ω be a bounded convex C^1 domain with $W(A) \subset \Omega$. Then*

$$\|p(A)\| \leq 2 \max_{z \in \Omega} |p(z)|. \quad (13.3)$$

Proof. First suppose B has simple spectrum and $W(B) \subset \Omega$. Set

$$m_\Omega = \max_{\Omega} |p|.$$

If $m_\Omega = 0$, the polynomial vanishes on the open set Ω , hence is identically zero. Otherwise let $f = p/m_\Omega$, so $\max_{\overline{\Omega}} |f| \leq 1$.

Use the nonexceptional perturbations above. Each $T_\ell = f_{\eta_\ell}(B)$ has simple spectrum and generates the same algebra as B . The auxiliary sharp bound [theorem 9.2](#) gives

$$\|f_{\eta_\ell}(B)\| \leq 2.$$

As $\eta_\ell \rightarrow 0$, $f_{\eta_\ell}(B) \rightarrow f(B)$. Therefore

$$\|p(B)\| = m_\Omega \|f(B)\| \leq 2m_\Omega.$$

Now choose simple-spectrum $B_k \rightarrow A$ with $W(B_k) \subset \Omega$. Polynomial evaluation and the operator norm are continuous, so passing $k \rightarrow \infty$ proves (13.3). $\square$

13.5 Smooth convex outer approximation

Lemma 13.3 (Parallel bodies). *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. For $\delta > 0$, define*

$$\Omega_\delta = \{z : \text{dist}(z, K) < \delta\}.$$

Then:

1. Ω_δ is a bounded convex domain containing K ;

2. $\partial\Omega_\delta$ is C^1 ;
3. $\overline{\Omega_\delta} = K + \delta\overline{\mathbb{D}}$, and

$$d_H(\overline{\Omega_\delta}, K) = \delta.$$

Proof. The identity $\overline{\Omega_\delta} = K + \delta\overline{\mathbb{D}}$ gives boundedness, convexity, and containment. It gives Hausdorff distance at most δ ; equality follows by taking a supporting point of K in any fixed direction and moving it outward by δ in that direction.

We prove the boundary regularity. Every $z \in \mathbb{C}$ has a unique nearest point $\pi_K(z) \in K$. Existence follows from compactness. For uniqueness, if two different minimizers existed, their midpoint would be in K and the parallelogram identity would make it strictly closer. To see continuity rather than assume it, let $p = \pi_K(z)$ and $q = \pi_K(w)$. Minimality along the line segments from p to q and from q to p gives the variational inequalities

$$\operatorname{Re}(\overline{z-p}(q-p)) \leq 0, \quad \operatorname{Re}(\overline{w-q}(p-q)) \leq 0.$$

Adding them yields $|p-q|^2 \leq \operatorname{Re}(\overline{z-w}(p-q))$, and Cauchy–Schwarz gives

$$|\pi_K(z) - \pi_K(w)| \leq |z - w|.$$

Thus the metric projection is nonexpansive, hence continuous.

Let $d_K(z) = \operatorname{dist}(z, K)$. Comparing $z+h$ first with $\pi_K(z)$, and z with $\pi_K(z+h)$, gives

$$\begin{aligned} 2 \operatorname{Re}(\overline{z+h - \pi_K(z+h)} h) - |h|^2 &\leq d_K(z+h)^2 - d_K(z)^2 \\ &\leq 2 \operatorname{Re}(\overline{z - \pi_K(z)} h) + |h|^2. \end{aligned}$$

Continuity of π_K shows that d_K^2 is continuously differentiable, with real gradient

$$\nabla d_K^2(z) = 2(z - \pi_K(z)). \quad (13.4)$$

On the level set $d_K(z) = \delta$, this gradient has length 2δ , so it never vanishes. Since

$$\partial\Omega_\delta = \{z : d_K(z)^2 = \delta^2\},$$

the implicit-function theorem gives a C^1 boundary. □

Intuition

Adding a small disk rounds every corner just enough for a continuous normal, without destroying convexity. The squared distance function is the defining equation, and its nonzero gradient is the outward normal direction.

Proposition 13.4 (Parallel-body boundary resolvent bound). *Let $B \in \mathbb{M}_n(\mathbb{C})$, put $K = W(B)$, and let*

$$\Omega_\delta = \{z : \operatorname{dist}(z, K) < \delta\}, \quad \delta > 0.$$

For $\zeta \in \partial\Omega_\delta$, write $\mathcal{R}_\zeta = (\zeta I - B)^{-1}$. Then

$$\int_{\partial\Omega_\delta} \mathcal{R}_\zeta^* \mathcal{R}_\zeta ds \preceq \frac{2\pi}{\delta} I. \quad (13.5)$$

Equivalently, for every $x \in \mathbb{C}^n$,

$$\int_{\partial\Omega_\delta} \|\mathcal{R}_\zeta x\|_2^2 ds \leq \frac{2\pi}{\delta} \|x\|_2^2.$$

The constant is sharp already when $B = 0$.

Proof. Fix $\zeta \in \partial\Omega_\delta$, put $p = \pi_K(\zeta)$, and use (13.4) to identify the outward normal:

$$\nu(\zeta) = \frac{\zeta - p}{\delta}.$$

The nearest-point variational inequality says that p is a support point of K in the direction $\nu(\zeta)$. Hence

$$\operatorname{Re}(\overline{\nu(\zeta)}p) = \lambda_{\max} \left(\operatorname{Re}(\overline{\nu(\zeta)}B) \right).$$

Since $\zeta = p + \delta\nu(\zeta)$, the support slack from theorem 8.1 is

$$\begin{aligned} D_{\Omega_\delta, B}(\nu(\zeta)) &= \delta I + \lambda_{\max} \left(\operatorname{Re}(\overline{\nu(\zeta)}B) \right) I - \operatorname{Re}(\overline{\nu(\zeta)}B) \\ &\succeq \delta I. \end{aligned}$$

Apply theorem 8.3:

$$2\pi I = \int_{\partial\Omega_\delta} \mathcal{R}_\zeta^* D_{\Omega_\delta, B}(\nu(\zeta)) \mathcal{R}_\zeta \, ds \succeq \delta \int_{\partial\Omega_\delta} \mathcal{R}_\zeta^* \mathcal{R}_\zeta \, ds.$$

For $B = 0$, direct integration on $\partial\Omega_\delta = \{|\zeta| = \delta\}$ gives equality. $\square$

Geometric smoothing is operator regularization

The support function of $W(B) + \delta\mathbb{D}$ is the support function of $W(B)$ plus δ . Thus adding the disk adds exactly δI to every directional support slack. The weighted conservation law from Week 8 then becomes the unweighted vector-valued resolvent estimate (13.5).

Common pitfall

Equation (13.5) controls the integral of $\mathcal{R}_\zeta^* \mathcal{R}_\zeta$ in operator order, equivalently one fixed vector at a time. It does not imply the dimension-free scalar inequality $\int \|\mathcal{R}_\zeta\|^2 \, ds \leq 2\pi/\delta$.

13.6 The limit ledger

There are three conceptually distinct limits:

$$\boxed{\eta_\ell \rightarrow 0 \quad \longrightarrow \quad B_k \rightarrow A \quad \longrightarrow \quad \overline{\Omega_{\delta_j}} \rightarrow W(A).} \quad (13.6)$$

1. $\eta_\ell \rightarrow 0$ removes simple spectrum and algebra generation for $f(B)$, while B itself still has simple spectrum.
2. $B_k \rightarrow A$ removes simple spectrum for the matrix.
3. $\delta_j \downarrow 0$ removes the smooth outer domain and returns to the exact numerical range.

Writing “by approximation” without this order hides which theorem is available at each stage.

13.7 Exercises

Exercise 13.1 (Warm-up: separate two values). Given distinct β_1, β_2 but equal values $f(\beta_1) = f(\beta_2)$, find all η for which $f(\beta_j) + \eta\beta_j$ still collide.

Exercise 13.2 (Core: exceptional parameters). For a simple-spectrum 3×3 diagonal B , write every exceptional value of η explicitly in terms of β_i and $f(\beta_i)$.

Exercise 13.3 (Core: algebra generation). Prove (13.2) in both directions, naming the polynomial used for each inclusion.

Exercise 13.4 (Core: fixed domain). Reconstruct theorem 13.2 and mark the exact line where each auxiliary hypothesis is used.

Exercise 13.5 (Core: projection gradient). Derive the two-sided inequality in the proof of theorem 13.3 and then prove (13.4).

Exercise 13.6 (Stretch: nonexpansive projection). Prove

$$|\pi_K(z) - \pi_K(w)| \leq |z - w|$$

using the variational inequalities for the two nearest points.

Chapter 14

Assembly of the main theorem and the spectral-set conclusion

Learning goals

- Reproduce the main polynomial proof in dependency order.
- Pass from outer-domain maxima to the numerical-range maximum.
- Pass the stronger orbit-Gramian certificate through all three approximation layers.
- Derive exact and quantitative rigidity for extremal pairs.
- Extend the polynomial estimate to rational functions.
- Distinguish theorem, classical input, and limiting step.

Workshop plan

Opening: place the ten proof dependencies in order. **Core:** shrink smooth outer domains and pass the polynomial bound to $W(A)$. **Guided proof:** use a pole-free neighborhood and Runge approximation for rational functions. **Studio:** submit the full proof write-up with every temporary hypothesis crossed out where it is removed.

14.1 The polynomial estimate

Proof of [theorem 4.3](#), polynomial part. Fix $A \in \mathbb{M}_n(\mathbb{C})$ and $p \in \mathbb{C}[z]$. The numerical range $K = W(A)$ is nonempty, compact, and convex. For $j \geq 1$, let

$$\Omega_j = \{z : \text{dist}(z, K) < 1/j\}.$$

By [theorem 13.3](#), these are bounded convex C^1 domains containing K , and

$$d_{\text{H}}(\overline{\Omega}_j, K) \rightarrow 0.$$

Apply the fixed-domain estimate:

$$\|p(A)\| \leq 2 \max_{z \in \overline{\Omega}_j} |p(z)|.$$

All the sets eventually lie in one compact neighborhood of K . Uniform continuity of $|p|$, or the Hausdorff-maxima exercise from Week 5, gives

$$\max_{z \in \Omega_j} |p(z)| \longrightarrow \max_{z \in W(A)} |p(z)|.$$

Pass to the limit:

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|.$$

□

14.2 A stronger orbit-Gramian conclusion

The approximation argument preserves more than the first-power norm bound. The canonical orbit Gramian from (12.12) passes through every limit even when the diagonalizing eigenbases themselves become singular.

Corollary 14.1 (Global orbit-Gramian estimate). *Let $p \in \mathbb{C}[z]$, and put*

$$m = \max_{z \in W(A)} |p(z)|.$$

If $m > 0$, define $T = p(A)/m$. Then, for every $0 < r < 1$,

$$\mathcal{P}_r(T) := \sum_{k=0}^{\infty} r^{2k} T^{*k} T^k \preceq \frac{1}{1-r} I. \quad (14.1)$$

In particular, at $r = 1/2$,

$$\sum_{k=0}^{\infty} 4^{-k} \|T^k x\|_2^2 \leq 2 \|x\|_2^2 \quad (x \in \mathbb{C}^n). \quad (14.2)$$

Proof. Fix $0 < r < 1$. At the auxiliary simple-spectrum, algebra-generating stage, Week 12 proved

$$\mathcal{P}_r(X) \preceq \frac{1}{1-r} I.$$

We record why this order bound survives the three limits in (13.6). For fixed N , put

$$S_N(X) = \sum_{k=0}^N r^{2k} X^{*k} X^k.$$

If $X_j \rightarrow X$ and $\mathcal{P}_r(X_j) \preceq (1-r)^{-1} I$, then polynomial continuity gives

$$S_N(X) = \lim_{j \rightarrow \infty} S_N(X_j) \preceq \frac{1}{1-r} I.$$

Apply this observation successively as the scalar perturbation tends to zero, the simple-spectrum matrices tend to A , and the outer domains shrink to $W(A)$. The matrices at the final outer-domain stage are

$$\frac{p(A)}{\max_{\Omega_j} |p|} \longrightarrow T.$$

Thus every finite partial sum $S_N(T)$ is bounded above by $(1-r)^{-1} I$. The partial sums increase in positive-semidefinite order, so they converge in finite dimensions and give (14.1). Equation (14.2) is its quadratic-form version at $r = 1/2$. □

Corollary 14.2 (Global controlled similarity). *With the notation of theorem 14.1, put $X_r = \mathcal{P}_r(T)^{1/2}$. Then*

$$\kappa(X_r) \leq \frac{1}{\sqrt{1-r}}, \quad \|X_r T X_r^{-1}\| \leq \frac{1}{\sqrt{r}}.$$

In particular, at $r = 1/2$, the normalized matrix T is similar, with condition number at most $\sqrt{2}$, to a matrix of norm at most $\sqrt{2}$.

Proof. Shifting the convergent series gives the Stein identity

$$\mathcal{P}_r(T) = I + r^2 T^* \mathcal{P}_r(T) T.$$

Now repeat the calculation following (12.13). The order bounds

$$I \preceq \mathcal{P}_r(T) \preceq (1-r)^{-1} I$$

control both the transformed norm and the condition number. $\square$

Remark 14.3. The raw Cauchy matrix P_r depends on a diagonalizing eigenbasis and need not converge when eigenvalues coalesce. The canonical object $\mathcal{P}_r(T)$ is basis-free; passing finite partial sums is what makes the strengthened conclusion stable. If $m = 0$, then the polynomial estimate just proved gives $p(A) = 0$, so there is no normalization to perform.

Corollary 14.4 (Nilpotent rigidity of exact equality). *Under the hypotheses of theorem 14.1, suppose*

$$\|p(A)\| = 2m.$$

Then there are orthonormal vectors $x, y \in \mathbb{C}^n$ such that

$$Tx = 2y, \quad Ty = 0.$$

Consequently, $\text{span}\{y, x\}$ is T -invariant and, in the ordered basis (y, x) ,

$$T|_{\text{span}\{y, x\}} = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}. \quad (14.3)$$

In particular, $p(A)$, and hence A , is not diagonalizable.

Proof. Choose a unit vector x with $\|Tx\|_2 = 2$, and let

$$P = \mathcal{P}_{1/2}(T).$$

By theorem 14.1,

$$2 \geq x^* P x = 1 + \frac{1}{4} \|Tx\|_2^2 + \sum_{k=2}^{\infty} 4^{-k} \|T^k x\|_2^2 \geq 2.$$

Thus $x^*(2I - P)x = 0$. Since $2I - P \succeq 0$, theorem 1.4 gives $Px = 2x$; the nonnegative tail also gives $T^2x = 0$. Put $y = Tx/2$. Then y is a unit vector and $Ty = 0$, so the Gramian series gives $Py = y$. Since P is Hermitian, its eigenvectors x and y , belonging to the distinct eigenvalues 2 and 1, are orthogonal. This proves (14.3).

The relations $T^2x = 0$ and $Tx \neq 0$ show that T is not diagonalizable. A polynomial of a diagonalizable matrix is diagonalizable, so A cannot be diagonalizable either. Also $0 \in \sigma(p(A))$, hence p vanishes at some eigenvalue of A . $\square$

Common pitfall

The two-dimensional subspace in (14.3) is invariant for T , but need not reduce T and need not be invariant for A . The corollary is a necessary condition for equality, not a classification of all extremal pairs.

Corollary 14.5 (Quantitative near-extremal chain). *Let $T = p(A)/m$ as above, and choose a unit vector x with $\|Tx\|_2 = \|T\|$. Then*

$$\left\|T^2x\right\|_2^2 \leq 4\left(4 - \|T\|^2\right). \quad (14.4)$$

Proof. The $k = 0$ and $k = 1$ terms in (14.2) give

$$\sum_{k=2}^{\infty} 4^{-k} \left\|T^kx\right\|_2^2 \leq 1 - \frac{1}{4} \|T\|^2 = \frac{4 - \|T\|^2}{4}.$$

Retain only the $k = 2$ term and multiply by 16. □

14.3 The proof in dependency order

Dependency-ordered reconstruction

1. Continuity on the finite-dimensional unit sphere makes $W(A)$ compact; Toeplitz–Hausdorff makes it convex.
2. A smooth convex outer domain supplies a positive double-layer map Φ .
3. The Cayley family gives a normalized positive-real H whose resolvent defect lies in $\text{alg}(T^*)$, provided $T = f(B)$ generates $\text{alg}(B)$.
4. Simple spectrum diagonalizes that algebraic defect.
5. Half-eigenvalue samples plus one origin sample cancel it.
6. The surviving kernel inequality compares two weighted Gramians.
7. Detector rigidity and the Stein identity yield $\|T\| \leq 2$.
8. The f_η perturbation enforces the two auxiliary hypotheses and then disappears.
9. Simple-spectrum matrices approach A inside the fixed domain.
10. Parallel outer domains shrink to $W(A)$.

14.4 From polynomials to rational functions

Let r be rational with no pole on $W(A)$. Its finite pole set is disjoint from the compact numerical range, so choose $\delta > 0$ such that the compact convex neighborhood

$$K_\delta = W(A) + \delta \overline{\mathbb{D}}$$

contains no pole. Then r is holomorphic on a neighborhood of K_δ , and $\mathbb{C} \setminus K_\delta$ is connected. Runge approximation gives polynomials p_j with

$$p_j \longrightarrow r \quad \text{uniformly on } K_\delta.$$

Choose a contour $\Gamma \subset \text{int } K_\delta$ enclosing $\sigma(A)$. Uniform convergence on Γ , together with the Cauchy functional calculus, gives

$$p_j(A) \longrightarrow r(A).$$

Uniform convergence on $W(A)$ also gives

$$\max_{W(A)} |p_j| \longrightarrow \max_{W(A)} |r|.$$

Apply the polynomial estimate and pass to the limit:

$$\|r(A)\| \leq 2 \max_{z \in W(A)} |r(z)|.$$

Thus $W(A)$ is a 2-spectral set for A .

Common pitfall

Uniform approximation only on $W(A)$ is not the stated reason that $p_j(A) \rightarrow r(A)$. The proof approximates on a pole-free *neighborhood* and uses convergence on a contour around the spectrum.

14.5 Proof-audit table

Temporary assumption	Why it appears	How it is removed
$\partial\Omega$ is C^1	Defines normal and double-layer density	Parallel bodies of $W(A)$
$W(A) \subset \Omega$	Keeps resolvents and boundary calculus valid	Hausdorff shrinkage
B has simple spectrum	Diagonal defect and Lagrange coordinates	Density plus $B_k \rightarrow A$
$f_\eta(B)$ has simple spectrum	Completion theorem hypothesis	Finite exceptional set, then $\eta \rightarrow 0$
$\text{alg}(f_\eta(B)) = \text{alg}(B)$	Places companion defect in $\text{alg}(T^*)$	Same separating perturbation
$ f_\eta \leq 1$	Cayley positivity	Normalization denominator

14.6 Exercises

Exercise 14.1 (Warm-up: shrinking disks). For $K = \overline{\mathbb{D}}$, write $\Omega_j = K + (1/j)\mathbb{D}$ explicitly and verify directly that $\max_{\overline{\Omega_j}} |z|^m \rightarrow \max_K |z|^m$.

Exercise 14.2 (Core: two-page proof). Write a self-contained two-page proof of the polynomial estimate, citing but not reproving the positive-real completion theorem and the double-layer lemma.

Exercise 14.3 (Core: maximum convergence). Give an ε -proof that

$$\max_{\overline{\Omega_j}} |p| \rightarrow \max_{W(A)} |p|.$$

Exercise 14.4 (Core: rational passage). Reproduce the rational approximation argument and explain why $\sigma(A) \subset W(A)$ is used.

Exercise 14.5 (Stretch: replace p). Show that the same argument proves

$$\|f(A)\| \leq 2 \max_{W(A)} |f|$$

for every function f holomorphic on a neighborhood of $W(A)$.

Chapter 15

Consequences, computation, and applications

Learning goals

- Apply the theorem to holomorphic matrix functions.
- Prove the Hilbert-space polynomial form by compression.
- Turn scalar approximation error into matrix approximation error.
- Design a numerical-range experiment without confusing evidence with proof.

Workshop plan

Opening: turn a scalar approximation error into a matrix error. **Core:** holomorphic and Hilbert-space consequences. **Computation lab:** sample support lines of a numerical range and calibrate on the exact nilpotent disk. **Closing:** separate numerical evidence from proof.

15.1 Holomorphic functions

Corollary 15.1 (Holomorphic form). *If f is holomorphic on a neighborhood of $W(A)$, then*

$$\|f(A)\| \leq 2 \max_{z \in W(A)} |f(z)|.$$

Proof. Choose a compact convex neighborhood of $W(A)$ on which f is holomorphic. Runge approximation gives polynomials converging uniformly there. Matrix evaluation converges by the Cauchy functional calculus, and the numerical-range maxima converge uniformly. $\square$

If s is a polynomial, or a rational function without poles on $W(A)$, apply the corollary to $f - s$:

$$\|f(A) - s(A)\| \leq 2 \max_{z \in W(A)} |f(z) - s(z)|. \quad (15.1)$$

This says that a good scalar approximant on the numerical range is, up to the sharp universal factor 2, a good matrix approximant.

15.2 Finite-dimensional compression on a Hilbert space

Let $\mathcal{H} \neq \{0\}$ be a complex Hilbert space, with inner product conjugate-linear in the first variable, and let

$$W(A) = \{\langle x, Ax \rangle : \|x\| = 1\}$$

for $A \in \mathcal{B}(\mathcal{H})$.

Corollary 15.2 (Hilbert-space polynomial form). *For every polynomial p ,*

$$\|p(A)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Consequently, $\overline{W(A)}$ is a 2-spectral set for A .

Proof. If $p = 0$, the claim is immediate. Otherwise let $d = \deg p$ and fix a unit vector x . Define the finite Krylov space

$$\mathcal{M}_x = \text{span} \{x, Ax, \dots, A^d x\}$$

and the compression

$$A_x = P_{\mathcal{M}_x} A|_{\mathcal{M}_x}.$$

Inductively,

$$A_x^k x = A^k x \quad (0 \leq k \leq d), \quad (15.2)$$

because $A^{k+1}x \in \mathcal{M}_x$ while $k < d$. Also $W(A_x) \subset W(A)$. The finite-dimensional theorem gives

$$\|p(A)x\| = \|p(A_x)x\| \leq \|p(A_x)\| \leq 2 \sup_{z \in W(A)} |p(z)|.$$

Take the supremum over unit x .

For the rational conclusion we use the standard inclusion

$$\sigma(A) \subseteq \overline{W(A)} \quad (15.3)$$

for a bounded operator on a complex Hilbert space. If a rational function r has no pole on $\overline{W(A)}$, compactness of its finite pole set gives a pole-free compact neighborhood of $\overline{W(A)}$. Approximate r there by polynomials and use a contour around $\sigma(A)$ inside that neighborhood. The polynomial bound and the Cauchy functional calculus then pass to the limit exactly as in Week 14. $\square$

Spectrum lies in the closed numerical range

If $\lambda \notin \overline{W(A)}$, a separating line and a rotation give $c > 0$ such that

$$\operatorname{Re} \langle x, e^{-i\theta}(A - \lambda I)x \rangle \geq c \|x\|^2$$

for every x . Cauchy–Schwarz shows that $A - \lambda I$ is bounded below. The analogous inequality for its adjoint shows that the adjoint has trivial kernel, so the range is both closed and dense, hence all of the Hilbert space. Therefore $A - \lambda I$ is invertible, proving (15.3).

15.3 A numerical-range plotting algorithm

Formula (3.2) suggests a robust boundary sampler:

1. choose angles $\theta_k = 2\pi k/N$;
2. form the Hermitian matrix

$$H_{\theta_k} = \operatorname{Re}(e^{-i\theta_k} A);$$

3. compute a normalized eigenvector x_k for its largest eigenvalue;
4. record $z_k = x_k^* A x_k$.

The line with outward normal $e^{i\theta_k}$ supports $W(A)$ at z_k . Repeated largest eigenvalues may produce flat faces, so one should sample the compression of $\operatorname{Im}(e^{-i\theta_k} A)$ to the top eigenspace if the entire face is desired.

15.4 A workshop experiment

For selected matrices and polynomials, approximate the Crouzeix ratio

$$\mathcal{C}(A, p) = \frac{\|p(A)\|}{\max_{z \in W(A)} |p(z)|}.$$

A careful experiment should include:

- a normal matrix, where the ratio is at most 1;
- the nilpotent sharp example, where the ratio equals 2;
- a mildly nonnormal triangular matrix;
- a random 3×3 matrix and several polynomial degrees;
- mesh-refinement checks for the denominator.

Numerical evidence can test intuition and reveal near-extremizers, but it cannot certify the universal inequality.

15.5 Why approximation bounds matter

Many matrix algorithms approximate $f(A)b$ by a polynomial or rational expression $s(A)b$. Equation (15.1) separates the work:

$$\text{scalar approximation on } W(A) \implies \text{matrix error control.}$$

This perspective is relevant to polynomial and rational Krylov methods, including GMRES; see the survey [3] and the numerical-range spectral-set literature.

15.6 Exercises

Exercise 15.1 (Core: holomorphic form). Prove theorem 15.1 with an explicit surrounding compact set and contour.

Exercise 15.2 (Core: Krylov agreement). Prove (15.2) by induction and verify $W(A_x) \subset W(A)$.

Exercise 15.3 (Core: error transfer). Derive (15.1) for both polynomial and rational s .

Exercise 15.4 (Project: numerical-range atlas). Implement the support-line sampler. Report the matrix, the sampled boundary, at least three polynomials, the observed ratios, and a mesh-refinement study. Include the exact nilpotent calculation as a calibration.

Exercise 15.5 (Stretch: flat faces). Choose a concrete 3×3 matrix for which the largest eigenvalue of H_θ has multiplicity two. Compute the tangential compression K_θ from [theorem 3.5](#) and use its extreme eigenvalues to decide whether the supported face is a point or a nontrivial segment. Give one example of each behavior.

Chapter 16

Scope, open directions, and a new-tool clinic

Learning goals

- Distinguish the scalar theorem from complete spectral-set questions.
- Separate the operator-valued support geometry of $W(A)$ from the domain-independent completion mechanism.
- Identify the hypotheses that make kernel cancellation work.
- Formulate plausible extensions without claiming them as theorems.
- Present the proof and its new tool coherently.

Workshop plan

Opening: distinguish scalar from complete spectral-set questions. **Core:** diagnose which hypotheses power the new tool and which are artifacts of this proof. **Capstone studio:** present one proof stage, one failure mode, and one plausible extension, labeling conjectures as conjectures.

16.1 What has been established

The course has developed the source manuscript's scalar statement:

$$\|p(A)\| \leq 2 \max_{z \in W(A)} |p(z)|.$$

It controls one scalar-valued polynomial at a time and yields the rational 2-spectral-set conclusion.

16.2 What is specifically special about the numerical range

The proof uses no curvature formula, conformal parametrization, or strict convexity of $W(A)$. For a bounded C^1 convex test domain Ω , define its support function on all of $\mathbb{T}$ by

$$h_{\Omega}(u) = \max_{z \in \overline{\Omega}} \operatorname{Re}(\overline{u}z).$$

When $\partial\Omega \cap \sigma(A) = \emptyset$, the numerical-range-specific bridge is

$$W(A) \subseteq \overline{\Omega} \iff h_{\Omega}(u)I - \operatorname{Re}(\bar{u}A) \succeq 0 \quad (u \in \mathbb{T}),$$

which resolvent congruence turns into the positive boundary density (8.3). In other words, every scalar supporting-half-plane inequality lifts exactly to operator order. The numerical range is the intersection, and the Hausdorff limit, of the smooth convex outer enclosures producing these positive kernels. This formulation allows $W(A)$ itself to have corners or to meet the spectrum on its boundary, where the boundary resolvent need not be defined.

After this bridge has produced the positive-real completion, the sharp argument is independent of the planar shape. The factor 2 comes from the optimally balanced sampling scale $r = 1/2$, the ordered Gramians, and the Stein identity. Thus the useful structure belongs to the pair consisting of the operator and its numerical range, not to the unlabeled convex set alone.

16.3 What is stronger and remains separate

For a matrix polynomial

$$\mathcal{F}(z) = [f_{ij}(z)] \in \mathbb{M}_m(\mathbb{C}[z]),$$

one can ask whether

$$\|\mathcal{F}(A)\| \leq 2 \max_{z \in W(A)} \|\mathcal{F}(z)\| \tag{16.1}$$

holds uniformly in m . Here $\mathcal{F}(A)$ acts on $(\mathbb{C}^n)^m$. The case $m = 1$ is the scalar theorem. Uniformity in m is an additional operator-algebraic property and does not follow formally from the scalar case; compare [8].

In the block-valued setting, a diagonal scalar nuisance becomes a family of noncommuting blocks. The single equation $Gv + Pu = 0$ no longer obviously annihilates every nuisance direction. A block or bimodule version of the cancellation principle would be needed.

Notice that the geometry-to-boundary-map step is already completely positive by the POVM construction in Week 8. The matrix-level obstruction is therefore not a failure of convex positivity. It is the algebraic de-symmetrization step: at matrix level the correction no longer reduces to scalar diagonal entries that one origin vector can cancel.

16.4 Potential extensions of the abstract tool

The positive-real completion theorem does not mention numerical ranges. This suggests several research programs:

- replace a finite diagonal algebra by a semisimple commutative algebra or a simultaneously diagonalizable tuple;
- find tangential or block annihilators for matrix-valued functions;
- seek forward–backward Gramian pairs for annular domains;
- replace a finite eigenbasis by spectral measures or a Riesz basis in infinite dimensions;
- classify equality beyond the necessary nilpotent restriction in [theorem 14.4](#), and sharpen the stability estimate (14.4).

These are directions, not consequences proved in this text.

16.5 Capstone options

1. Give a line-by-line expository proof of [theorem 9.1](#), with a symbolic 2×2 audit.
2. Derive the general-radius estimate ([12.11](#)) and explain its optimization geometrically.
3. Compute the double-layer measure on a disk and compare it with the Poisson kernel.
4. Starting from [theorem 14.4](#), investigate which additional conditions make the invariant nilpotent restriction sufficient for equality.
5. Compare scalar spectral sets with the matrix-level question ([16.1](#)).
6. Design an abstract nuisance module and determine whether a finite anchor sample can annihilate it.

16.6 Final conceptual summary

The idea to remember

The earlier symmetrized boundary identity contains a useful companion term but does not isolate $f(B)$. The source proof retains the full Cayley family, revealing that the resolvent defect lies in an adjoint-generated algebra. Simple spectrum turns that algebraic fact into a diagonal analytic correction. Herglotz-kernel samples at half the conjugate eigenvalues, together with a compensating sample at the origin, eliminate the correction exactly. The remaining inequality orders two weighted Gramians. Their positive difference detects the dynamics, and the narrower Gramian satisfies a Stein identity. That combination forces the sharp norm bound 2.

The proof also leaves behind more structure than the norm estimate: the support slacks resolve the identity through the boundary resolvents, the canonical orbit Gramians survive all approximation layers, and exact saturation forces the normalized functional-calculus output

$$T = \frac{p(A)}{\max_{z \in W(A)} |p(z)|}$$

to contain an invariant 2×2 nilpotent restriction.

16.7 Research-status note

The repository describes the source as a *candidate proof* for which formal peer review is pending. It also records computational, adversarial, and formalization audits that found no specific error. Accordingly, this workshop presents the manuscript's argument in a checkable form and preserves the distinction between a research manuscript, formal verification in a stated system, and independent mathematical peer review. The additional support-slack, orbit-Gramian, and equality-rigidity results in these notes are presented as deductions from the manuscript's mechanism, not as claims of priority in the literature.

16.8 Exercises

Exercise 16.1 (Core: scalar versus complete). Explain why applying the scalar theorem to each entry of $\mathcal{F}$ does not prove (16.1) with constant 2.

Exercise 16.2 (Core: tool diagnosis). For a proposed extension, identify: the positive kernel, the nuisance module, the annihilator equation, the two ordered Gramians, and the Stein recursion. If one component is missing, state the obstruction.

Exercise 16.3 (Capstone oral). Give a fifteen-minute proof overview in which every classical input, new step, and approximation step is explicitly labeled.

Exercise 16.4 (Stretch: stability). Suppose the final norm is close to 2. Trace which inequalities must be nearly equalities and formulate a quantitative stability question.

Appendix A

Proof dependencies and equation ledger

A.1 Minimal dependency graph

The main proof can be checked in four independent packets.

Packet A: finite-dimensional linear algebra

- simple spectrum implies diagonalizability;
- Lagrange interpolation identifies generated algebras;
- congruence preserves positivity;
- the polar decomposition balances a nonorthogonal eigenbasis;
- Hermitian eigenvectors test matrix inequalities.

Packet B: analytic positivity

- scalar Herglotz representation;
- polarization to a positive matrix-valued measure;
- positivity of the matrix Herglotz kernel;
- scalar Cayley transform maps the disk to the right half-plane.

Packet C: numerical-range boundary calculus

- Toeplitz–Hausdorff convexity and $\sigma(B) \subset W(B)$;
- supporting-line geometry and its operator-valued support slacks;
- the positive double-layer density, its exact containment test, and mass $2I$;
- the support–resolvent resolution of the identity and the parallel-body L^2 resolvent bound;
- the boundary POVM and symmetrized normal dilation;
- the companion identity;
- smooth parallel outer domains.

Packet D: the new mechanism

- the Cayley family makes a resolvent completion;
- the defect lies in $\text{alg}(T^*)$;
- simple spectrum makes the defect diagonal after congruence;
- half-eigenvalue and origin samples cancel it;
- two Gramians, detector rigidity, and a Stein identity give 2;
- arbitrary-radius sampling gives a family of canonical orbit-Gramian certificates;
- those certificates survive approximation and force nilpotent rigidity at equality.

Packets A–C are classical ingredients. Packet D is the manuscript’s central new mechanism.

A.2 Normalization ledger

Quantity	Where it comes from	What it does
$2I$	Total mass of two boundary layers	Makes $\Phi = \frac{1}{2} \int d\mu_B$ unital
2	Cayley expansion $1 + 2 \sum_{m \geq 1} w^m f^m$	Turns the known double-layer coefficient into T^m
$1/2$	Sample $w_i = \bar{\lambda}_i/2$	Produces the paired weights $1/2$ and $1/4$
P	Kernel denominator $1 - w_i \bar{w}_j$	Determines the compensator $v = -G^{-1}Pu$
Q	Resolvent denominator $1 - w_i \lambda_j$	Creates the wider Gramian
$4Q - 2P$	Partial fractions in the sample block	Leads to the coefficient 4 in the anticommutator
4	Detector–anticommutator coefficient	Gives $\hat{P} \preceq 2I$
$1/4$	Stein recursion weight	Turns $\hat{P} \preceq 2I$ into $\tilde{T}^* \tilde{T} \preceq 4I$

A.3 One-page completion proof

Assume $T = S\Lambda S^{-1}$, $G = S^*S$, and

$$\tilde{H}(w) = G(I - w\Lambda)^{-1} + \Theta(w)G, \quad \Theta(0) = 0, \quad \Theta(w) \text{ diagonal.}$$

Set

$$P_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j/4}, \quad Q_{ij} = \frac{G_{ij}}{1 - \bar{\lambda}_i \lambda_j/2}, \quad Y = Q - P.$$

Sample the positive Herglotz kernel at

$$w_i = \bar{\lambda}_i/2, \quad \xi_i = u_i e_i, \quad w_0 = 0, \quad \xi_0 = -G^{-1}Pu.$$

The correction contribution contains $Gv + Pu = 0$, hence vanishes. The known blocks are $4Q - 2P$, $G + Q$, and $2G$, so

$$4Y - YG^{-1}P - PG^{-1}Y \succeq 0.$$

After balancing,

$$4\hat{Y} - \hat{Y}\hat{P} - \hat{P}\hat{Y} \succeq 0,$$

with

$$\hat{P} = \sum_{k \geq 0} 4^{-k} \tilde{T}^{*k} \tilde{T}^k, \quad \hat{Y} = \sum_{k \geq 1} (2^{-k} - 4^{-k}) \tilde{T}^{*k} \tilde{T}^k \succeq 0.$$

The detector lemma gives $I \preceq \hat{P} \preceq 2I$. The Stein identity

$$\hat{P} = I + \frac{1}{4} \tilde{T}^* \hat{P} \tilde{T}$$

then gives

$$\tilde{T}^* \tilde{T} \preceq 4(\hat{P} - I) \preceq 4I.$$

Because T and $\tilde{T}$ are unitarily equivalent, $\|T\| \leq 2$.

Appendix B

Hints and selected solutions

B.1 Weeks 1–4

The sharp matrix norm ([exercise 1.2](#))

$$N^*N = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}.$$

Its largest eigenvalue is 4, so $\|N\| = 2$.

The 2×2 PSD test ([exercise 1.3](#))

Necessity of $a, d \geq 0$ follows by testing the coordinate vectors. If $a > 0$, complete the square:

$$\begin{pmatrix} \bar{x} & \bar{y} \end{pmatrix} H \begin{pmatrix} x \\ y \end{pmatrix} = a \left| x + \frac{b}{a}y \right|^2 + \left(d - \frac{|b|^2}{a} \right) |y|^2.$$

This is nonnegative exactly when $ad \geq |b|^2$. If $a = 0$, the determinant condition forces $b = 0$, and $d \geq 0$ finishes.

Interpolation ([exercise 2.2](#))

For distinct λ_j , set

$$\ell_j(z) = \prod_{k \neq j} \frac{z - \lambda_k}{\lambda_j - \lambda_k}, \quad q(z) = \sum_j d_j \ell_j(z).$$

Then $q(\lambda_j) = d_j$. Substituting a diagonalization of T gives $q(T) = S \operatorname{diag}(d_j) S^{-1}$.

Equality of generated algebras ([exercise 2.3](#))

$f(B) \in \operatorname{alg}(B)$ gives $\operatorname{alg}(f(B)) \subseteq \operatorname{alg}(B)$. If the $f(\beta_i)$ are distinct, choose a polynomial q with $q(f(\beta_i)) = \beta_i$. Since B is diagonalizable, $q(f(B)) = B$, so the reverse inclusion holds.

The nilpotent disk ([exercise 3.2](#))

Up to a global phase, every unit vector is $x = (\cos t, e^{i\phi} \sin t)^T$. Then

$$x^* N x = e^{i\phi} \sin(2t).$$

The radius $\sin(2t)$ takes every value in $[0, 1]$, and ϕ takes every angle. Hence the image is exactly $\overline{\mathbb{D}}$.

B.2 Weeks 5–8

Matrix Cauchy mass (exercise 5.2)

If $B = SAS^{-1}$, integrate

$$(\zeta I - B)^{-1} = S \operatorname{diag}((\zeta - \lambda_j)^{-1}) S^{-1}$$

entrywise. Each diagonal contour integral is 1, so the result is $SIS^{-1} = I$. For a general matrix, apply the holomorphic functional calculus to the constant function 1, or use Jordan form and note that higher-order pole terms integrate to zero.

Herglotz factorization (exercise 7.2)

Use $\bar{\zeta} = \zeta^{-1}$ and

$$\frac{\overline{\zeta + z}}{\zeta - z} = \frac{1 + \bar{z}\zeta}{1 - \bar{z}\zeta}.$$

Bring the two fractions to a common denominator. The numerator becomes $2(1 - w\bar{z})$, which cancels the outer factor.

Double-layer density (exercise 8.2)

Let $R = Z^{-1}$. Then

$$R^*(\nu Z^* + \bar{\nu} Z)R = \nu R^* Z^* R + \bar{\nu} R^* Z R = \nu R + \bar{\nu} R^*.$$

The middle expression is positive by congruence.

Companion identity (exercise 8.4)

Insert the boundary formula for $g_h(z)$ into its Cauchy functional calculus on an interior contour Γ . Interchange the two finite contour integrals. For fixed boundary ζ , the inner integral is

$$\frac{1}{2\pi i} \int_{\Gamma} \frac{(zI - B)^{-1}}{\zeta - z} dz = (\zeta I - B)^{-1}.$$

The contour remains strictly inside Ω , so no boundary extension of g_h is required.

B.3 Weeks 9–12

Cayley coefficient audit (exercise 9.3)

$$H(w) = \Phi(1) + 2 \sum_{m \geq 1} w^m \Phi(f^m) = I + \sum_{m \geq 1} w^m (T^m + g_{f^m}(B)^*).$$

The first equality uses bounded linearity of Φ ; the second uses the double-layer identity once for every m .

Wrong algebra orientation (exercise 9.6)

Since $\rho(wT_M) < 1$,

$$I - (I - wT_M)^{-1} = - \sum_{k \geq 1} w^k T_M^k \in \text{alg}(T_M).$$

But

$$\|T_M\| \geq \|T_M e_2\|_2 = \sqrt{M^2 + \frac{1}{4}} \rightarrow \infty.$$

Thus the conclusion fails with $\text{alg}(T)$.

Correction cancellation (exercise 10.4)

The half of the kernel quadratic form containing $E(w_i)$ is

$$\sum_{i=1}^n (u_i e_i)^* E(w_i) \left(v + \sum_{j=1}^n \frac{u_j e_j}{1 - w_i \overline{w_j}} \right).$$

Since $E(w_i) = \Theta(w_i)G$, this is the expression inside the real part in (10.8). The adjoint half is its complex conjugate. The denominator creates P , so the bracket is $(Gv + Pu)_i = 0$.

Known top-left block (exercise 11.2)

For $F(w) = G(I - w\Lambda)^{-1}$,

$$F(w)_{ij} = \frac{G_{ij}}{1 - \overline{\lambda_i} \lambda_j / 2},$$

and $F(w_j)_{ij}^*$ is the same. Divide their sum by $1 - w_i \overline{w_j} = 1 - \overline{\lambda_i} \lambda_j / 4$ and apply the stated partial fraction identity.

Gramian expansion (exercise 11.5)

$$P = \sum_{k \geq 0} 4^{-k} \Lambda^{*k} G \Lambda^k.$$

Congruence by $G^{-1/2}$ changes the k -th summand into

$$G^{-1/2} \Lambda^{*k} G \Lambda^k G^{-1/2} = \tilde{T}^{*k} \tilde{T}^k.$$

The calculation for Q is identical with 2^{-k} .

Stein shift (exercise 12.3)

Remove the $k = 0$ term from $\hat{P}$, factor one $\tilde{T}^*$ on the left and one $\tilde{T}$ on the right, and reindex. Then

$$\tilde{T}^* \tilde{T} \preceq \tilde{T}^* \hat{P} \tilde{T} = 4(\hat{P} - I) \preceq 4I.$$

General radius (exercise 12.4)

Solving

$$\frac{2}{(1-rx)(1-r^2x)} = \frac{a}{1-rx} + \frac{b}{1-r^2x}$$

gives $a = 2/(1-r)$ and $b = -2r/(1-r)$. After the same graph substitution, pure P_r -terms cancel because $a(1-r) = 2$. Rigidity gives $\hat{P}_r \preceq (1-r)^{-1}I$, and the Stein identity yields $\|\tilde{T}\|^2 \leq [r(1-r)]^{-1}$. Differentiate $r(1-r)$, or complete the square:

$$r(1-r) = \frac{1}{4} - (r - \frac{1}{2})^2.$$

B.4 Weeks 13–16**Exceptional parameters (exercise 13.2)**

For each $i < j$, a collision occurs only at

$$\eta_{ij} = -\frac{f(\beta_i) - f(\beta_j)}{\beta_i - \beta_j}.$$

The common positive denominator in f_η does not affect collisions.

Projection gradient (exercise 13.5)

For the upper bound compare $z + h$ with the candidate $\pi_K(z)$:

$$d_K(z + h)^2 \leq |z + h - \pi_K(z)|^2.$$

Expand and subtract $d_K(z)^2$. For the lower bound compare z with $\pi_K(z + h)$ and rearrange. Divide the resulting two-sided estimate by $|h|$; continuity of π_K leaves the real-linear derivative

$$h \mapsto 2 \operatorname{Re}(\overline{z - \pi_K(z)}h).$$

Two-page proof outline (exercise 14.2)

Use exactly four paragraphs:

1. fixed Ω , simple-spectrum approximation B_k ;
2. normalize p , perturb f_η , and obtain algebra generation;
3. invoke the double-layer completion and completion theorem, then take $\eta \rightarrow 0$ and $B_k \rightarrow A$;
4. take parallel outer domains to $W(A)$.

Any proof that changes this order should explicitly justify why the required hypotheses are still available.

Rational passage (exercise 14.4)

Choose a pole-free compact convex neighborhood K_δ , use Runge on that set, and choose a contour around $\sigma(A)$. The inclusion $\sigma(A) \subset W(A)$ ensures that the contour can be placed inside the pole-free neighborhood and that the rational functional calculus is defined.

Krylov compression (exercise 15.2)

Assume $A_x^k x = A^k x$ for $k < d$. Then

$$A_x^{k+1} x = P_{\mathcal{M}_x} A A^k x = A^{k+1} x,$$

because $A^{k+1} x \in \mathcal{M}_x$. For a unit $y \in \mathcal{M}_x$,

$$\langle y, A_x y \rangle = \langle y, P_{\mathcal{M}_x} A y \rangle = \langle y, A y \rangle,$$

so $W(A_x) \subset W(A)$.

Appendix C

Glossary and notation

Symbol or term	Meaning
$\mathbb{C}, \mathbb{D}, \mathbb{T}$	Complex plane, open unit disk, and unit circle
$\mathbb{M}_n(\mathbb{C})$	$n \times n$ complex matrices
A^*	Conjugate transpose or Hilbert-space adjoint
$\ x\ _2$	Euclidean vector norm
$\ A\ $	Induced Euclidean operator norm
$\sigma(A)$	Spectrum, the set of eigenvalues in finite dimensions
$W(A)$	Numerical range $\{x^*Ax : \ x\ _2 = 1\}$
$\operatorname{Re} A$	Hermitian part $(A + A^*)/2$
$H \succeq 0$	H is positive semidefinite
$H \preceq K$	$K - H \succeq 0$
$\operatorname{alg}(T)$	Unital algebra of polynomials in T
C -spectral set	Compact $K \supset \sigma(A)$ satisfying $\ r(A)\ \leq C \max_K r $
resolvent	$(zI - A)^{-1}$ when $z \notin \sigma(A)$
simple spectrum	n distinct eigenvalues for an $n \times n$ matrix
congruence	Transformation $H \mapsto S^*HS$, preserving positivity
similarity	Transformation $A \mapsto S^{-1}AS$, preserving spectrum
Carathéodory function	Analytic H on $\mathbb{D}$ with $\operatorname{Re} H \succeq 0$
positive kernel	Kernel satisfying every finite sampled quadratic inequality (7.1)
Φ	Unital positive double-layer map
POVM	Positive operator-valued measure of total mass I
g_h	Holomorphic Cauchy companion of boundary data $\bar{h}$
$D_{\Omega, B}(u)$	Directional support-slack matrix $h_{\Omega}(u)I - \operatorname{Re}(\bar{u}B)$
$G = S^*S$	Gram matrix of a diagonalizing eigenbasis
P, Q	Cauchy-type matrices that become two weighted Gramians
$Y = Q - P$	Positive detector after balancing
$\mathcal{P}_r(T)$	Canonical orbit Gramian $\sum_{k \geq 0} r^{2k} T^{*k} T^k$
Stein identity	Discrete recursion $P = I + aT^*PT$
d_H	Hausdorff distance between compact sets

Appendix D

Classical inputs: exact statements for review

D.1 Toeplitz–Hausdorff

For every bounded operator A on a complex Hilbert space, its numerical range is convex. In finite dimensions it is also compact. The course proves the finite-dimensional case by 2×2 compression in Week 3.

D.2 Scalar Herglotz

An analytic scalar function on $\mathbb{D}$ has nonnegative real part if and only if it has the representation

$$f(w) = ic + \int_{\mathbb{T}} \frac{\zeta + w}{\zeta - w} d\mu(\zeta),$$

where $c \in \mathbb{R}$ and μ is a finite positive measure. The representation is unique. Week 7 derives the matrix form and the kernel consequence.

D.3 Naimark dilation for a POVM

Let E be a positive operator-valued measure on a compact space X , acting on a Hilbert space $\mathcal{H}$, and suppose $E(X) = I_{\mathcal{H}}$. Then there are a Hilbert space $\mathcal{K}$, an isometry $V : \mathcal{H} \rightarrow \mathcal{K}$, and a projection-valued measure F on X such that

$$E(\mathcal{E}) = V^* F(\mathcal{E}) V$$

for every Borel set $\mathcal{E} \subseteq X$. Equivalently, the unital positive map

$$\varphi \mapsto \int_X \varphi dE$$

is the compression of a unital $*$ -representation of $C(X)$, and is therefore completely positive.

A proof constructs $\mathcal{K}$ by completing vector-valued simple functions modulo the nullspace of the semidefinite form

$$[f, g]_E = \int_X f(x)^* dE(x) g(x).$$

Multiplication by characteristic functions gives the projections $F(\mathcal{E})$, and constant functions define the isometry V . Week 8 applies this construction with $X = \partial\Omega$.

D.4 Runge

If $K \subset \mathbb{C}$ is compact with connected complement, every function holomorphic near K is uniformly approximable on K by polynomials. The course applies this only to compact convex sets.

D.5 Implicit-function theorem

If a C^1 real-valued function on $\mathbb{R}^2$ has nonzero gradient on a level set, that level set is locally a C^1 curve. Week 13 applies this to $d_K^2(z) = \delta^2$.

D.6 Finite-dimensional polar decomposition

Every invertible complex matrix has a unique factorization

$$S = U(S^*S)^{1/2},$$

where U is unitary. This converts the nonunitary diagonalization into the unitary equivalence $T = U(G^{1/2}\Lambda G^{-1/2})U^*$.

Provenance and attribution

These course notes are based on the local source manuscript *The numerical range is a 2-spectral set* by Shanmu Jin. The source manuscript states that OpenAI ChatGPT contributed the idea of sampling the matrix Herglotz kernel at half the conjugate eigenvalues and adding the origin sample that cancels the adjoint-algebra correction, as well as assisting with exposition, bibliography, and typesetting. The source assigns responsibility for the mathematical statements and citations to its human author.

This workshop edition expands the definitions, prerequisites, proof calculations, exercises, approximation ledger, and the general-radius interpretation. It also records further deductions from the proof: the support-slack boundary calculus, its resolvent conservation law, the canonical orbit-Gramian estimate, and nilpotent rigidity at equality. These additions should be read together with the research-status note in Week 16.

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